

SISURAKSHA: INTELLIGENT CHILD SAFETY PLATFORM FOR SCHOOL BUSES

Group Number: 25-26J-282

BSc (Hons) degree in Information Technology
Specializing in Information Technology

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DECLARATION OF THE CANDIDATE AND SUPERVISOR

I declare that this is my own work and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

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ABSTRACT

School transportation safety remains a major operational and social concern in Sri Lanka due to the continued reliance on fragmented manual procedures, limited safety monitoring mechanisms, and inefficient transportation management practices. Existing school bus systems primarily focus on basic tracking and administrative functionality while lacking proactive accident prevention, tamper-resistant attendance verification, intelligent emergency response, and dynamic route optimization capabilities. This research presents SISURAKSHA, an integrated intelligent child safety platform designed to improve school transportation safety, accountability, and operational efficiency through the integration of emerging digital technologies.

The proposed platform combines four interoperable subsystems within a unified cyber-physical architecture. These include a blockchain-based RFID attendance management system with secure hash-chain verification, a multi-sensor crash and hazard detection framework, a movement-aware footboard safety monitoring system, and an Edge-AI biometric authentication and emergency management subsystem. The system integrates IoT edge computing, machine learning, computer vision, blockchain verification, and intelligent routing technologies to provide real-time monitoring and automated decision-making capabilities.

Experimental evaluation demonstrated high operational reliability across all subsystems. The RFID attendance module achieved a 100% scan success rate, while the Edge-AI biometric system achieved 99.12% identification accuracy with low inference latency. The accident detection subsystem successfully identified multiple hazard conditions using sensor fusion, and the attendance-driven routing engine reduced unnecessary transportation route lengths by up to 11.4%.

The research concludes that integrated intelligent transportation ecosystems can significantly enhance child safety, operational transparency, emergency responsiveness, and transportation efficiency within school transportation environments. The proposed platform provides a scalable foundation for future smart transportation infrastructure development in Sri Lanka and similar developing regions.

Keywords: School Bus Safety, IoT, Blockchain, Machine Learning, YOLOv8, Footboard Safety

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LIST OF ABBREVIATIONS

Abbreviation	Description
ADDR	Attendance-Driven Dynamic Routing
AES	Advanced Encryption Standard
API	Application Programming Interface
CA-AEES	Context-Aware Automated Emergency Egress System
CNN	Convolutional Neural Network
CPS	Cyber-Physical System
DSRM	Design Science Research Methodology
Edge-AI	Edge Artificial Intelligence
ESP32	Espressif Systems 32-bit Microcontroller
GPS	Global Positioning System
GSM	Global System for Mobile Communications
IoT	Internet of Things
IR	Infrared
ML	Machine Learning
MTCNN	Multi-task Cascaded Convolutional Networks
OSRM	Open Source Routing Machine
PDPA	Personal Data Protection Act (Sri Lanka)
RFID	Radio Frequency Identification
SHA-256	Secure Hash Algorithm 256-bit
SLIIT	Sri Lanka Institute of Information Technology
SMS	Short Message Service
YOLO	You Only Look Once

CHAPTER 1: INTRODUCTION

1.1 General Introduction

School transportation constitutes a critical component of the educational and social infrastructure serving thousands of families across Sri Lanka on a daily basis. For many students, particularly those residing in suburban and rural regions, school buses represent the primary means of access to education. Consequently, the safety, reliability, and operational efficiency of school transportation systems directly influence not only the wellbeing and security of children, but also the confidence and peace of mind of parents, guardians, educational institutions, and transport providers. A safe and dependable transportation environment contributes significantly to consistent school attendance, reduced parental anxiety, and the overall sustainability of educational operations. Conversely, failures within transportation systems can lead to severe physical, emotional, financial, and institutional consequences affecting multiple stakeholders simultaneously.

Despite the critical importance of this domain, school transportation systems in Sri Lanka continue to depend heavily on manual, fragmented, and technologically outdated operational practices. Attendance tracking within many school bus services is still conducted using handwritten registers, verbal confirmations, or isolated digital spreadsheets that lack real-time synchronization and verification capabilities. Similarly, route planning and student allocation processes are often managed manually by drivers or fleet coordinators without intelligent optimization mechanisms. Safety monitoring practices remain largely reactive, relying on human observation rather than automated detection systems capable of identifying hazards in real time. These limitations create substantial vulnerabilities in accident prevention, emergency response coordination, attendance accountability, and operational transparency.

The tragic Kotmale bus accident of May 2025, which resulted in multiple fatalities involving school-aged passengers, starkly exemplifies the catastrophic consequences of systemic weaknesses in existing school transportation infrastructure [1], [2]. Incidents of this nature highlight the urgent need for proactive technological interventions capable of improving situational awareness, accelerating emergency response, and minimizing preventable risks within school transport ecosystems. The event also underscored broader concerns regarding vehicle monitoring, passenger accountability, and emergency management preparedness in Sri Lanka's public and private school transportation sectors.

According to the World Health Organization's Global Status Report on Road Safety 2023, road traffic injuries remain one of the leading causes of mortality globally, particularly among children and young adults, while Sri Lanka records a disproportionately high annual road accident fatality rate relative to its vehicle population density [1], [3]. School transportation environments are especially vulnerable due to the concentration of minor passengers within confined and mobile operational spaces. Although the Sri Lankan regulatory framework governing school transportation mandates measures such as semi-annual vehicle inspections, speed governors, driver licensing requirements, and the visual standardization of school bus identification, these mechanisms primarily function as passive compliance-oriented controls. While such regulations are essential, they are fundamentally reactive in nature and provide limited capability for real-time intervention during active emergency scenarios.

In practical operational contexts, the limitations of existing systems become increasingly evident during critical incidents. For example, in the event of vehicle rollovers, fires, flooding, or submersion-related emergencies, manual emergency exits may become inaccessible within seconds due to structural deformation, panic-induced congestion, or hydraulic pressure accumulation [4], [5]. Similarly, conventional GPS tracking systems commonly deployed in school buses merely provide location visibility without offering intelligent hazard analysis, automated emergency escalation, or behavioral monitoring

capabilities. The absence of integrated sensor fusion technologies and autonomous alerting mechanisms significantly delays emergency response times, thereby increasing the severity of accident outcomes.

Beyond physical safety concerns, conventional school transportation ecosystems also suffer from major economic, administrative, and logistical inefficiencies. One of the most persistent operational problems is the reliance on static enrollment manifests that do not dynamically reflect daily attendance variations. As a result, drivers are frequently required to visit every registered pickup location regardless of whether the respective student is actually attending school on a given day. This phenomenon, commonly referred to as the “ghost stop problem,” leads to substantial unnecessary fuel consumption, increased travel durations, vehicle wear, and inefficient route utilization [6], [31]. Over time, these inefficiencies contribute to elevated operational costs for private transport providers and schools while simultaneously increasing traffic congestion and environmental emissions.

Furthermore, existing attendance management systems often lack verifiable and tamper-resistant recordkeeping mechanisms. Traditional centralized databases and paper-based records are vulnerable to unauthorized modification, accidental data loss, and synchronization inconsistencies. Such vulnerabilities reduce transparency and accountability across the transportation ecosystem, particularly during disputes involving student boarding verification, route compliance, or emergency investigations. In environments involving minors, these deficiencies represent not only operational weaknesses but also significant trust and governance concerns.

Another growing challenge within modern school transportation systems involves the secure and ethical handling of sensitive student data. With the increasing adoption of biometric technologies and digital identity systems, concerns surrounding privacy protection, unauthorized data sharing, and regulatory compliance have become increasingly significant. Many existing cloud-dependent systems transmit sensitive

student information to external servers, creating potential exposure risks and raising compliance concerns under emerging data protection regulations such as Sri Lanka's Personal Data Protection Act (PDPA) No. 9 of 2022 [38]. Consequently, there exists a pressing requirement for decentralized and privacy-preserving architectures capable of performing intelligent processing directly at the network edge without excessive dependence on cloud infrastructure.

Within this context, SISURAKSHA, the Smart School Bus Safety and Monitoring Ecosystem, was conceived as a comprehensive and integrated technological solution designed to address the multidimensional shortcomings of existing school transportation systems. The platform aims to unify attendance management, real-time safety monitoring, intelligent route optimization, biometric verification, hazard detection, and emergency response coordination within a single cohesive cyber-physical ecosystem. Rather than functioning as a standalone application or isolated monitoring tool, SISURAKSHA is architecture as a modular and interoperable platform capable of supporting multiple intelligent subsystems operating collaboratively in real time.

SISURAKSHA was developed as a group research project and capstone academic contribution by four undergraduate computing students at the Sri Lanka Institute of Information Technology (SLIIT). The ecosystem adopts a distributed architectural approach in which each team member designed and implemented an independent but interoperable subsystem targeting a distinct research gap within the school transportation domain. These subsystems collectively integrate technologies including Internet of Things (IoT) edge devices, blockchain verification frameworks, machine learning-based computer vision, biometric authentication, intelligent routing algorithms, and multi-sensor emergency detection mechanisms. Through this multidisciplinary integration, the SISURAKSHA platform seeks to transform conventional school transportation from a reactive and manually managed environment into a proactive, intelligent, and data-driven child safety ecosystem capable of improving operational accountability,

emergency responsiveness, and transportation efficiency across the Sri Lankan educational sector.

1.2 The SISURAKSHA Ecosystem

The SISURAKSHA platform comprises four principal subsystems specifically designed for integration within a unified and interoperable school bus operational environment. Each subsystem addresses a distinct yet interconnected category of challenges associated with student safety, transportation accountability, operational intelligence, and emergency management. While each component functions independently with its own dedicated hardware, software, and analytical mechanisms, the subsystems are collectively integrated through a shared cyber-physical architecture that enables coordinated decision-making and real-time information exchange. This modular design approach improves scalability, maintainability, and deployment flexibility while ensuring that all safety-critical functions contribute toward a single cohesive intelligent transportation ecosystem.

The architecture of SISURAKSHA emphasizes proactive intervention rather than passive monitoring. Traditional school transportation systems typically rely on isolated technologies that address only individual operational concerns such as GPS tracking or attendance logging. In contrast, SISURAKSHA combines blockchain verification, Internet of Things (IoT) sensing, machine learning inference, Edge-AI processing, and intelligent routing algorithms into a synchronized platform capable of detecting, analyzing, and responding to real-world transportation events in real time. The integration of these technologies enables the ecosystem to provide enhanced student accountability, rapid emergency response coordination, operational transparency, and fuel-efficient transportation management within the Sri Lankan school transportation context.

The first subsystem, titled the Blockchain Based RFID Attendance Management System and Segmented Polyline Analysis for Verifiable Driver-Student Matchmaking, was

developed by M. H. S. Akther (IT22590312). This subsystem serves as the trust and attendance intelligence layer of the SISURAKSHA ecosystem. The component integrates RFID-based student attendance verification with a hybrid blockchain validation framework designed to ensure tamper-resistant attendance records. By combining SHA-256 cryptographic hash chaining with off-chain data persistence, the subsystem provides verifiable integrity without incurring the high transaction costs associated with fully on chain architectures. In addition to attendance management, the subsystem incorporates segmented polyline analysis using OSRM road-network routing to improve the accuracy of driver-student assignment and route validation processes. The subsystem also adopts an offline-first IoT architecture utilizing MicroSD buffering and deferred synchronization to maintain operational continuity during network outages.

The second subsystem, the Smart Bus Crash Detection and Safety Window Monitoring System, was developed by T. A. P. K. Shameera (IT22606624). This subsystem focuses primarily on accident prevention, hazard detection, and emergency escalation within school transportation environments. The architecture utilizes an ESP32 microcontroller integrated with a multi-sensor array capable of detecting rollover events, collisions, vibration anomalies, smoke, fire, gas leaks, water hazards, and abnormal pressure conditions. To minimize false emergency alerts, the subsystem incorporates a driver verification and cancellation mechanism that allows controlled suppression of accidental triggers within a predefined verification window. Additionally, the subsystem integrates a YOLOv8-based computer vision model for real-time safety window monitoring, enabling automated detection of unsafe passenger behavior such as excessive leaning or dangerous positioning near bus windows. Through the integration of machine learning and sensor fusion technologies, the subsystem enhances situational awareness and supports faster emergency response coordination.

The third subsystem, titled the Machine Learning and IoT-Based Footboard Safety and Driver Monitoring System, was developed by P. R. I. S. R. Pinto (IT22610102). This subsystem addresses one of the highest-risk physical zones in school transportation

operations: the vehicle entrance and footboard area. The architecture combines infrared proximity sensing with Hall effect pulse-based movement detection to identify unsafe passenger occupancy conditions during vehicle motion. Unlike conventional GPS-based speed estimation systems, the Hall effect sensor mechanism enables reliable movement classification through propeller shaft rotational analysis, reducing dependence on external satellite connectivity. The subsystem implements a three-state safety classification model consisting of SAFE, WARNING, and CRITICAL operational states, allowing dynamic risk evaluation based on passenger proximity and vehicle movement status. This movement-aware monitoring capability significantly improves the detection of dangerous boarding and alighting conditions while supporting proactive driver intervention.

The fourth subsystem, the Smart Access and Emergency Management System with Edge-AI Facial Recognition and Centralized Emergency Control, was developed by A. M. D. C. Kumara (IT22600134). This subsystem focuses on biometric accountability, privacy-preserving student verification, and intelligent emergency management. The architecture employs a fine-tuned MobileFaceNet convolutional neural network combined with an MTCNN facial detection pipeline to perform real-time student identification directly on edge-computing hardware. By executing biometric inference locally on devices such as the NVIDIA Jetson Nano, the subsystem minimizes cloud dependency and improves compliance with Sri Lanka's Personal Data Protection Act (PDPA) regulations governing biometric data processing. In addition to biometric verification, the subsystem incorporates a Context-Aware Automated Emergency Egress System (CA-AEES) and a centralized emergency control mechanism designed to support rapid evacuation and coordinated emergency response actions during critical incidents. The subsystem also introduces an Attendance-Driven Dynamic Routing engine capable of optimizing transportation routes based on real-time student attendance data, thereby reducing unnecessary travel distances, fuel wastage, and operational inefficiencies associated with ghost-stop routing.

Collectively, these four subsystems form the foundation of the SISURAKSHA ecosystem and demonstrate the feasibility of a unified, intelligent, and technology-driven approach to school transportation safety and management. By integrating blockchain security, IoT sensing, machine learning analytics, Edge-AI inference, and dynamic routing intelligence into a single interoperable architecture, SISURAKSHA addresses multiple previously unresolved research gaps while establishing a scalable framework for future smart transportation systems in Sri Lanka.

1.3 Background and Literature Survey

1.3.1 RFID-Based Attendance Systems in School Transportation

RFID (Radio Frequency Identification) technology has been extensively researched and implemented across educational institutions, transportation systems, logistics environments, and access-control infrastructures for automated identification and attendance management purposes. Its affordability, reliability, low power requirements, and ease of integration with embedded systems have made RFID one of the most practical technologies for large-scale operational environments involving repetitive identity verification tasks [7]. Within school transportation systems, RFID-based attendance mechanisms provide a more efficient alternative to traditional manual attendance registers by enabling automated student boarding and departure verification in real time [9].

Several previous studies have demonstrated the effectiveness of RFID-based attendance monitoring in reducing human error, improving attendance accuracy, and increasing parental visibility into student transportation activities [9]-[11]. Existing implementations commonly utilize passive RFID tags assigned to students and RFID readers installed within school buses to record boarding events automatically. These systems are often integrated with GSM or Wi-Fi communication modules to transmit attendance notifications to parents or centralized school management systems. Such

approaches improve accountability and reduce the administrative burden placed on drivers and school staff [10], [11].

Despite these advantages, existing RFID attendance systems still exhibit several critical limitations. Most currently deployed solutions rely entirely on centralized database architectures that lack tamper-resistance and cryptographic verification capabilities. Attendance records stored within centralized systems remain vulnerable to unauthorized modification, accidental deletion, synchronization inconsistencies, and insider manipulation [25]-[29]. Furthermore, many IoT-based attendance systems depend heavily on uninterrupted internet connectivity, causing data loss or synchronization failures during network outages [10], [11]. Existing implementations also rarely incorporate intelligent route optimization or geospatial verification mechanisms capable of validating whether assigned drivers and students are geographically compatible within real-world road networks [31]-[34].

Recent advancements in blockchain architectures and distributed ledger technologies have created opportunities for more secure attendance management frameworks. Blockchain-based educational systems have demonstrated significant potential in ensuring immutability, transparency, and decentralized verification of academic and operational records [25], [26]. However, existing blockchain attendance implementations often suffer from scalability limitations and high operational costs when large volumes of real-time attendance data are stored directly on-chain [27]-[29]. These limitations highlight the need for hybrid architectures capable of balancing cryptographic integrity with operational efficiency.

Consequently, there remains a significant research opportunity for the development of an integrated attendance ecosystem capable of combining RFID automation, blockchain verification, offline-first IoT persistence, and intelligent route optimization within a unified school transportation platform tailored to the Sri Lankan context.

1.3.2 Blockchain-Based Verification Systems for Educational Applications

Blockchain technology has emerged as a transformative approach for ensuring data integrity, decentralization, transparency, and tamper resistance across various sectors including finance, healthcare, logistics, and education [25]. In educational environments, blockchain frameworks have increasingly been explored for secure certificate validation, attendance verification, student identity management, and decentralized academic recordkeeping [26], [28]. The core architectural strength of blockchain lies in its immutable ledger structure, where records are cryptographically linked through hashing algorithms, making unauthorized modification computationally difficult and immediately detectable.

Research conducted by Grech and Camilleri demonstrated the potential of blockchain technologies to improve trust and accountability in educational systems through decentralized digital records [25]. Similarly, Chen et al. highlighted the applicability of blockchain architectures for secure educational data management and distributed verification frameworks [26]. More recent attendance-oriented blockchain systems have attempted to integrate smart contracts and distributed ledger validation to reduce fraud and enhance transparency in academic environments [27].

Despite these benefits, existing blockchain attendance systems continue to face major practical limitations in transportation-related deployments. Public blockchain infrastructures often introduce high transaction fees, confirmation delays, and scalability bottlenecks when processing large volumes of frequent attendance events [25], [27]-[29]. Additionally, most existing blockchain attendance research focuses primarily on institutional classroom environments rather than dynamic transportation ecosystems involving mobile IoT devices and intermittent connectivity conditions.

Another major limitation is the absence of hybrid architectures capable of balancing operational efficiency with blockchain-level integrity verification. Fully decentralized storage approaches remain economically impractical for real-time transportation systems

generating continuous attendance events throughout daily operations. Furthermore, many existing systems do not integrate route intelligence, geospatial analysis, or offline synchronization capabilities, limiting their practicality for school transportation environments [31]-[34].

Therefore, there remains a clear research need for a hybrid blockchain architecture capable of providing tamper-resistant attendance verification, scalable off-chain persistence, low operational cost, and offline-first IoT compatibility within school transportation ecosystems. The SISURAKSHA platform addresses this gap through a SHA-256-based hybrid blockchain framework integrated with off-chain MongoDB storage and Ethereum hash anchoring mechanisms.

1.3.3 IoT-Based Accident Detection and Emergency Monitoring Systems

The application of Internet of Things (IoT) technologies in transportation safety has increased significantly due to advances in low-cost sensors, embedded microcontrollers, wireless communication protocols, and real-time data processing capabilities [12]-[14]. IoT-based safety systems enable continuous environmental monitoring and autonomous event detection through interconnected sensor networks capable of collecting and transmitting operational data in real time.

Several smart transportation safety systems proposed in previous studies utilize basic accelerometers, vibration sensors, GPS modules, and legacy GSM communication technologies for accident detection and emergency notification [4], [16]-[18]. Such systems improve emergency responsiveness by automatically notifying predefined contacts or authorities following accident events. However, most existing implementations remain limited in functionality and rely heavily on single-sensor triggering mechanisms that are highly susceptible to false positives caused by potholes, rough road topography, or sudden non-critical braking conditions.

Additionally, many existing accident monitoring systems fail to incorporate comprehensive environmental and situational hazard detection capabilities. Critical post-accident hazards such as smoke accumulation, toxic gas leakage, water intrusion or vehicle submersion, fire outbreaks, and severe structural pressure impacts are often excluded from existing implementations despite their critical importance during transit emergencies [4], [5], [24]. Furthermore, conventional systems generally depend on manual passenger communication or driver intervention to verify an incident, which may become functionally impossible during severe high-impact collisions, driver incapacitation, or vehicle submersion scenarios.

Modern embedded platforms, such as the ESP32 microcontroller, have enabled more sophisticated multi-sensor fusion architectures capable of integrating multi-axial accelerometric, structural, and environmental hazard-monitoring sensors within a single, cost-effective edge-compute device [12]-[14], [24]. Multi-sensor fusion approaches materially improve telemetry reliability by correlating multiple independent sensor readings—such as cross-referencing localized vibration spikes with force-sensitive resistor (FSR) pressure thresholds before triggering emergency alerts. This collaborative mapping drastically reduces false-positive rates while significantly improving real-time situational awareness.

Despite these advancements, there remains limited research specifically focused on school transportation safety frameworks capable of combining multi-modal accident detection, environmental hazard monitoring, cellular-based remote escalation, and driver-in-the-loop false-alert suppression within a unified, cohesive architecture. The SISURAKSHA platform addresses this core research gap through an ESP32-based multi-sensor accident and hazard detection subsystem. By combining multi-modal edge sensor fusion with an automated 60-second driver verification countdown state machine, the system optimizes emergency management execution while systematically minimizing false-positive emergency broadcasts.

1.3.4 Computer Vision-Based Window Safety and Student Monitoring Systems

The integration of computer vision and deep learning technologies within intelligent transportation systems (ITS) has expanded significantly, driven by their capacity to perform automated, real-time spatial and behavioral analysis. Recent breakthroughs in convolutional neural networks (CNNs) and single-stage object detection frameworks, most notably the YOLO (You Only Look Once) architecture family, have transitioned visual recognition from resource-heavy cloud networks to localized, real-time edge-computing safety domains.

Modern iteration paradigms, specifically the YOLOv8 object detection model, offer key architectural enhancements, including anchor-free detection mechanics, decoupled heads, and highly optimized feature extraction pipelines. These engineering advancements drastically reduce inference latency while maintaining high bounding box precision. Consequently, these models are uniquely suited for deployment on decentralized edge processing platforms, providing the high-frequency frame rate processing required to analyze continuous, dynamic micro-behaviors inside moving school transit environments.

Despite these rapid technological developments, contemporary student transportation services heavily lack automated visual monitoring mechanisms to enforce safe boundary limits around open bus windows. Pervasive, high-risk student behaviors such as students extending their hands, arms, or heads beyond the window perimeter remain entirely dependent on intermittent manual supervision by the driver or a transit assistant. This introduces profound operational vulnerabilities, as the vehicle operator cannot maintain strict operational focus on complex road conditions while simultaneously monitoring multiple student windows.

Furthermore, existing computer vision research within the transit domain is heavily skewed toward external Advanced Driver Assistance Systems (ADAS) or internal driver drowsiness tracking, leaving a distinct research void in proactive student behavioral boundary compliance. The few systems that do attempt interior monitoring frequently rely on cloud-tethered processing pipelines. These architectures introduce fatal transmission latency bottlenecks, demand continuous broadband network availability, and introduce severe data privacy and ethical concerns regarding the streaming of minors' biometric data to external servers.

Therefore, a critical research gap persists for fully localized, edge-compatible vision systems capable of performing real-time spatial boundary-violation analysis within the bus cabin. The SISURAKSHA ecosystem addresses this gap through its dedicated Vision-Based Window Safety and Student Monitoring Subsystem. Powered by an onboard Raspberry Pi 5 edge compute core linked to a dual-camera input framework, the system runs a highly specialized YOLOv8 inference engine. By continuously tracking the window boundary threshold at the edge, the system instantly identifies high-risk anatomical exposure and triggers an immediate localized audio warning sequence via a dedicated digital media controller and speaker, dynamically mitigating student injuries before structural or vehicle impacts occur.

1.3.5 Footboard Safety and Vehicle Movement Monitoring Systems

The footboard and entrance region of school buses represent one of the highest-risk zones within student transportation environments. Accidents involving students boarding or exiting moving vehicles continue to pose serious safety concerns in many transportation systems, particularly in developing countries where overcrowding and inadequate monitoring practices remain common.

Traditional transportation safety systems primarily focus on external vehicle tracking and often neglect localized passenger risk analysis around the footboard region. Existing implementations generally lack movement-aware safety classification capable of

distinguishing between safe stationary occupancy and dangerous occupancy during vehicle motion.

Many current transportation systems rely heavily on GPS-based movement estimation for operational monitoring. While GPS provides useful positional information, it suffers from several practical limitations including signal instability, reduced accuracy in obstructed environments, latency issues, and dependence on external satellite communication [18]. These limitations reduce the reliability of GPS-only movement classification approaches in real-time safety-critical applications.

Hall effect sensors provide an alternative approach for movement detection through magnetic field variation analysis and rotational pulse monitoring [23]. Such sensors are widely utilized in automotive and industrial systems due to their reliability, low power consumption, and environmental resilience. However, their application within school transportation footboard safety systems remains relatively underexplored in existing literature.

Furthermore, most existing transportation safety systems provide only binary occupancy detection outputs without contextual movement-based risk evaluation. This creates operational limitations because occupancy near a stationary bus entrance may be harmless while identical occupancy during vehicle movement becomes critically dangerous.

Consequently, there remains a need for movement-aware footboard safety systems capable of integrating occupancy detection with reliable motion classification mechanisms. The SISURAKSHA platform addresses this gap through a Hall effect pulse-based movement detection subsystem integrated with infrared proximity sensors to provide dynamic SAFE, WARNING, and CRITICAL risk classification during school bus operations.

1.3.6 Edge AI Facial Recognition and Privacy Compliant Biometric Systems

Biometric identification technologies have become increasingly prevalent across transportation, healthcare, security, and educational environments due to their ability to provide reliable identity verification without requiring physical tokens or passwords. Facial recognition systems in particular have gained widespread adoption because of their contactless operation, automation capability, and suitability for real-time access-control environments.

Recent advances in lightweight convolutional neural network architectures such as MobileFaceNet have enabled efficient biometric inference on resource-constrained edge devices while maintaining high identification accuracy and low latency. Similarly, Multi-task Cascaded Convolutional Networks (MTCNN) provide robust face detection and preprocessing suitable for real-time embedded deployment. These technologies create opportunities for privacy-preserving Edge-AI biometric systems capable of operating independently from cloud infrastructure.

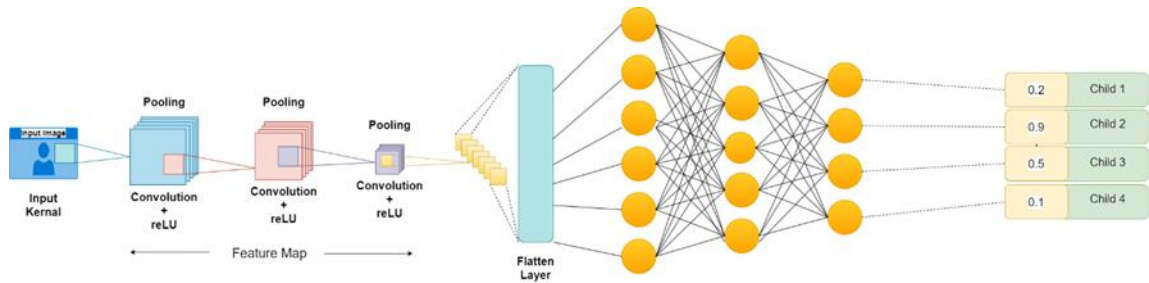


Figure 1.1: Review of architecture techniques for Edge-AI Convolutional Neural Networks

However, many currently deployed biometric verification systems rely heavily on centralized cloud processing architectures. Such approaches introduce several limitations including increased latency, network dependency, bandwidth consumption, and privacy risks associated with transmitting sensitive biometric data to external servers. These concerns become particularly critical within school transportation systems involving minors and sensitive personal information.

Sri Lanka's Personal Data Protection Act (PDPA) No. 9 of 2022 establishes important legal and ethical obligations regarding the collection, processing, and storage of biometric information [38]. Existing cloud-centric biometric systems may struggle to satisfy strict privacy compliance requirements due to external data transmission and third-party storage exposure risks. Furthermore, many large-scale biometric systems prioritize computational complexity and accuracy at the expense of edge-device deployment feasibility.

Additionally, conventional school transportation systems rarely integrate biometric verification with emergency management, route optimization, or attendance-driven operational intelligence. This fragmentation reduces overall system interoperability and operational efficiency.

Therefore, there remains a strong research requirement for decentralized Edge-AI biometric architectures capable of performing secure offline inference directly on local hardware while maintaining high accuracy, low latency, and regulatory compliance. The SISURAKSHA platform addresses this gap through a MobileFaceNet-based Edge-AI facial recognition subsystem deployed on embedded edge hardware with localized inference, encrypted biometric storage, and Attendance-Driven Dynamic Routing integration for intelligent school transportation management [38], [40].

1.3.7 Graph Theory and Dynamic Routing Optimization

An interesting benefit of graph theory in computational mathematics is that road networks can be modeled as dynamic datasets. The Attendance-Driven Dynamic Routing (ADDR) module refers to a technique where predictive modeling on student absences can be used to accelerate and optimize the physical trajectory of the bus. In this domain, the road network is modeled as a weighted directed graph $G = (V, E)$, where V represents the bus stops (vertices) and E represents the roads (edges).

Following is the general outline for the dynamic routing optimization used in the SSBMS:

1. Load the static daily enrollment manifest (the baseline directed graph).
2. Ingest proactive absence data submitted by guardians via the mobile application.
3. Automatically prune (remove) the unneeded vertices (absent student stops) from the graph.
4. Calculate the new optimal path using the standard edge traversal cost function.

Fine-tune the route by applying an International Roughness Index (IRI) penalty derived from the vehicle's onboard accelerometers to avoid heavily degraded roads

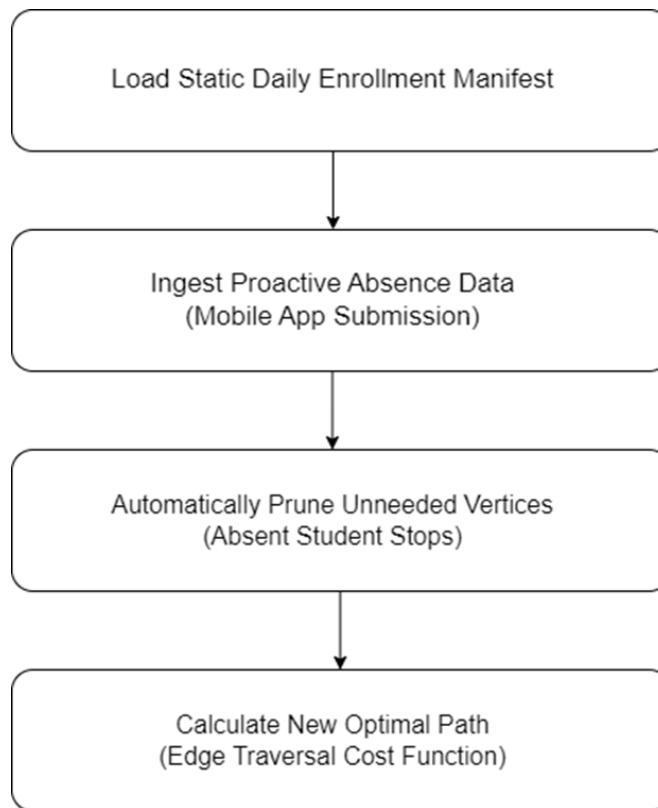


Figure 1.2: Dynamic Routing Optimization

1.3.8 Dual-Verification Failsafe Interlocks

As mentioned in many transportation safety research papers, automated mechanisms are currently the main tool used for rapid emergency response. Although there are great achievements, vehicular automation has some relevant challenges to tackle. The most frequently mentioned problem is the introduction of secondary hazards (e.g., an automated door deploying into active oncoming traffic). One of the ways of dealing with this problem is through multi-layered hardware interlocks. For ensuring absolute safety, the SSBMS relies on a Dual-Verification Kinematic and Perimeter Interlock. As an automated door requires absolute certainty before actuation, various environmental checks are applied to the data pipeline to guarantee safe deployment.

The operational logic of the interlock mechanism, as depicted in the flowchart, executes through a strict sequential state machine:

1. Hazard Detection (Trigger Phase): The interlock sequence is initiated either autonomously via the sensor fusion pipeline (detecting critical thresholds of fire, water submersion, or toxic gas leaks inside the cabin) or manually via an emergency push-button triggered by the driver. Once a hazard is confirmed, the system enters the verification phase.
2. Verification 1 (Kinematic Stasis Check): Before any mechanical release occurs, the microcontroller queries the onboard Hall-effect sensors. The logic gate requires the host vehicle's speed to be strictly 0 km/h. If the bus is still moving, the actuation command is queued and placed in a holding state until the vehicle comes to a complete halt.
3. Verification 2 (Dynamic Perimeter Check): Once the bus achieves 0 km/h, the front and rear-facing proximity sensors scan the external environment. Crucially, logic is designed to filter out static environment data; it ignores stationary objects (such as parked vehicles, trees, or walls). The interlock specifically scans for moving vehicles approaching the egress zone. If an active, moving threat is

detected, the door remains temporarily locked to prevent passengers from evacuating directly into oncoming traffic.

4. Actuation Phase: Only when the hazard trigger is active, Verification 1 confirms zero velocity, and Verification 2 confirms no moving vehicles in the perimeter, does the control logic return a Boolean TRUE. The system then immediately triggers the solenoid to deploy the emergency door automatically, allowing safe egress.

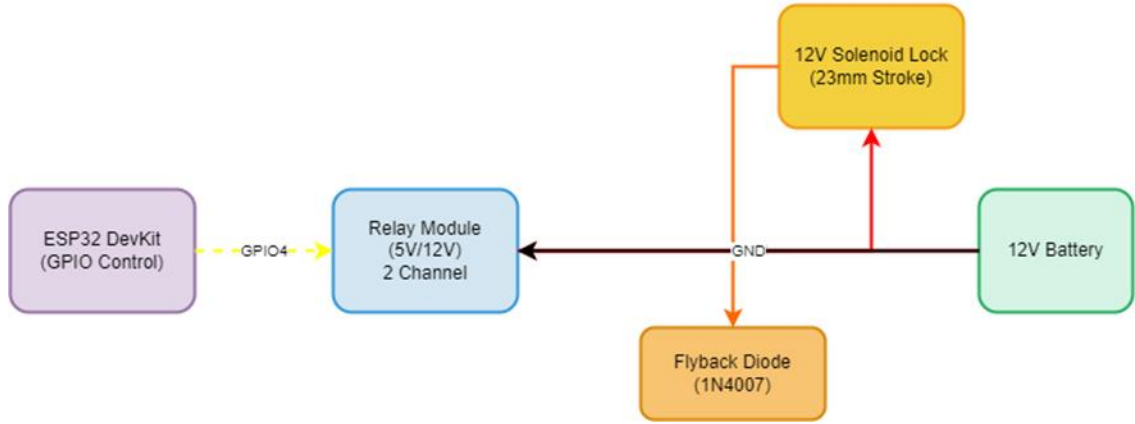


Figure 1.3: Sensor Fusion and Interlock Logic Flowchart.

System / Product	Technology	Attendance Method	Blockchain?	Route Intelligence	Key Limitation
Manual roll call	Paper-based	Manual	No	None	Error-prone; no real-time visibility
Barcode / QR scanner systems	Barcode / QR	Scan on boarding	No	None	No tamper-proof records; fragile codes
GPS tracking only	GPS + GSM	Location only	No	Straight-line	No attendance verification
RFID + GSM systems	RFID + SMS	RFID scan	No	None	SMS latency; no integrity guarantee
Hyperledger	RFID + HLF	RFID +	Yes	None	High latency;

Fabric attendance		blockchain	(permissioned)		complex deployment
Ethereum full on-chain attendance	RFID + Ethereum	RFID + on-chain	Yes	None	Prohibitive gas costs
Proposed SISURAKSH A module	RFID + hybrid blockchain + polyline	RFID + IoT + chain	Yes (hybrid)	Segmented polyline	Novel contribution of this research

Table 1.1: Existing School Transportation Technologies and Their Limitations

1.4 Research Gap

Based on the literature survey and comprehensive evaluation of existing school transportation technologies, several critical research gaps were identified across attendance accountability, transportation intelligence, accident prevention, passenger safety monitoring, biometric privacy, and route optimization domains. These gaps collectively demonstrate the absence of a unified, proactive, and intelligent school transportation safety ecosystem capable of addressing both operational efficiency and child safety requirements simultaneously.

Gap 1: Absence of Tamper-Proof Attendance Records in School Transportation Systems

Most existing RFID-based attendance systems rely on centralized databases that are vulnerable to unauthorized modification, accidental deletion, and synchronization inconsistencies. These systems do not provide cryptographic integrity guarantees capable of verifying whether attendance records have been altered after creation [25], [26]. As a result, transportation stakeholders lack a trustworthy and verifiable attendance validation mechanism. This research gap is addressed by Akther through the

implementation of a hybrid blockchain architecture utilizing SHA-256 hash chaining to provide tamper-evident attendance verification.

Gap 2: Prohibitive Cost of Full On-Chain Blockchain Storage for High-Frequency Attendance Data

While blockchain technology offers strong immutability and transparency properties, storing complete attendance records directly on public blockchain networks introduces substantial transaction costs, scalability limitations, and performance bottlenecks [28], [29]. Existing educational blockchain implementations often fail to achieve practical deployment feasibility due to excessive operational expenses. This research gap is addressed by Akther through a hybrid architecture combining off-chain MongoDB persistence with hash-only Ethereum anchoring to maintain cryptographic verification while significantly reducing blockchain transaction overhead.

Gap 3: IoT Offline Vulnerability Leading to Data Loss During Network Outages

Many IoT attendance and transportation monitoring systems depend heavily on uninterrupted internet connectivity for real-time data transmission. In practical transportation environments, temporary network disruptions frequently occur due to poor signal coverage, especially in rural or geographically obstructed areas [10], [11], [30]. Existing systems often fail to preserve data integrity during such outages, resulting in attendance loss and synchronization inconsistencies. This gap is addressed by Akther through the implementation of a MicroSD-based offline-first architecture with deferred synchronization capabilities enabling reliable local persistence during connectivity interruptions.

Gap 4: Inaccuracy of Straight-Line Distance Metrics in Driver-Student Route Matching

Conventional transportation assignment systems commonly utilize simplistic straight-line or Euclidean distance calculations when matching students to drivers or bus routes. Such approaches fail to reflect real-world road network constraints, traffic patterns, and route accessibility conditions, leading to inefficient route planning and inaccurate transportation assignments [31]-[34]. This gap is addressed by Akther through segmented polyline analysis integrated with OSRM road-network routing to improve driver-student matching accuracy based on realistic travel paths.

Gap 5: Absence of Automated Multi-Sensor Accident and Hazard Detection in School Buses

Existing school transportation systems largely depend on manual human observation for accident recognition and hazard response. Most currently deployed monitoring solutions lack integrated multi-sensor architectures capable of autonomously detecting rollover events, collisions, smoke, fire, gas leaks, water hazards, or abnormal pressure conditions in real time [4], [16], [17], [24]. This absence significantly delays emergency response during critical incidents. This research gap is addressed by Shameera through the development of a multi-sensor ESP32-based accident and hazard detection system incorporating a 60-second driver verification mechanism to reduce false emergency alerts.

Gap 6: Lack of Real-Time Unsafe Window Behavior Detection Using Computer Vision

Current school transportation monitoring systems rarely incorporate computer vision technologies for passenger behavioral analysis. Unsafe window-related behavior such as excessive leaning, dangerous positioning, or partial body exposure often remains undetected until accidents occur. Existing solutions generally lack automated visual

monitoring capable of continuously evaluating passenger safety conditions inside moving buses. Recent advances in object detection models such as YOLO have demonstrated the feasibility of real-time behavioral analysis in embedded environments. However, these technologies remain underutilized in school transportation safety applications. This gap is addressed by Shameera through the implementation of a YOLOv8-based machine learning model for real-time safety window behavior detection and monitoring.

Gap 7: Absence of Movement-Aware Footboard Occupancy Risk Classification

The footboard region of school buses represents one of the highest-risk physical safety zones, particularly during boarding and alighting operations. Existing transportation safety systems generally fail to distinguish between safe stationary occupancy and dangerous occupancy occurring while the vehicle is in motion. Furthermore, many systems rely exclusively on GPS-based movement estimation, which may be unreliable in certain operational environments [18], [22], [23]. This gap is addressed by Pinto through the development of a movement-aware footboard monitoring system combining Hall effect pulse-based movement detection with infrared proximity sensing for dynamic SAFE, WARNING, and CRITICAL risk classification.

Gap 8: Privacy Compliant Biometric Boarding Verification for School Bus Environments

Biometric attendance and facial recognition systems increasingly depend on cloud-based inference architectures that transmit sensitive student data to external servers for processing. Such approaches introduce privacy concerns and potential non-compliance with Sri Lanka's Personal Data Protection Act (PDPA) regulations governing biometric data processing for minors [38], [40], [42]. Existing systems also struggle to achieve low-latency inference performance on edge devices. This research gap is addressed by Kumara through the implementation of an Edge-AI Mobile FaceNet facial recognition

system capable of performing offline biometric inference directly on local edge hardware while maintaining PDPA compliance.

Gap 9: Ghost Stop Fuel Wastage Caused by Static Routing Without Dynamic Absence Integration

Traditional school transportation routes are generally designed using static enrollment data without considering daily attendance variations. Consequently, drivers continue visiting pickup locations even when registered students are absent, leading to unnecessary travel distances, fuel wastage, increased operational costs, and environmental inefficiency [6], [31], [43], [44]. Existing routing systems rarely incorporate real-time attendance-driven route optimization mechanisms. This gap is addressed by Kumara through the development of an Attendance-Driven Dynamic Routing engine utilizing graph pruning algorithms to eliminate unnecessary stops and optimize route efficiency dynamically.

Gap No.	Research Gap	Addressed By
Gap 1	Absence of tamper-proof attendance records	Hybrid blockchain with SHA-256 hash chaining and Ethereum hash anchoring
Gap 2	Prohibitive full on-chain blockchain storage cost	Off-chain storage with hash-only blockchain anchoring
Gap 3	IoT offline vulnerability during network outages	MicroSD offline-first buffering with deferred synchronization
Gap 4	Inaccuracy of straight-line distance in driver-student matching	OSRM road-network routes and segmented polyline analysis
Gap 5	Absence of automated multi-sensor accident and hazard detection	ESP32-based sensor fusion for rollover, vibration, impact, pressure, fire, smoke, gas and water hazards
Gap 6	Lack of real-time unsafe window behavior detection	YOLOv8-based computer vision and local audio warning
Gap 7	Absence of movement-aware	IR occupancy sensing plus Hall effect pulse-

	footboard occupancy risk classification	based movement detection
Gap 8	Privacy-compliant biometric boarding verification	Edge-AI MobileFaceNet and MTCNN with local inference and encrypted biometric templates
Gap 9	Ghost-stop fuel wastage caused by static routing	Attendance-Driven Dynamic Routing with graph pruning for absent students

Table 1.2: Research Gap Analysis and Addressing Mechanisms

1.5 Research Problem

Contemporary school transportation systems continue to face significant limitations due to the absence of a unified, secure, and geospatially intelligent management framework. Attendance records stored in conventional centralized databases lack cryptographic integrity guarantees, leaving them vulnerable to unauthorized modification and tampering. IoT-based attendance devices frequently lose data during network outages due to inadequate offline-first capabilities. Route assignment systems relying on simplistic straight-line distance calculations produce suboptimal driver-student matches that fail to reflect real-world road conditions. Safety monitoring remains dependent on manual observation, creating critical delays in hazard detection and emergency response. The sensitive biometric data of minor students is frequently transmitted to external cloud servers in violation of applicable data protection regulations. No existing system integrates all of these dimensions into a cohesive, modular, and practically deployable cyber-physical platform.

1.6 Research Objectives

The SISURAKSHA group research project pursues the following collective objectives:

- To design, implement, and evaluate an IoT Smart Attendance Device with offline-first data persistence, hybrid blockchain verification, and segmented polyline analysis for intelligent driver-student route matchmaking.

- To develop a multi-sensor accident and emergency detection system with false-alert suppression and a YOLOv8-based machine learning module for real-time unsafe window behavior monitoring.
- To implement a movement-aware footboard safety monitoring system using Hall effect pulse detection and infrared proximity sensing for critical risk classification during bus operations.
- To design and evaluate an Edge-AI biometric attendance verification system using a fine-tuned MobileFaceNet CNN deployed on an edge device, combined with an Attendance-Driven Dynamic Routing engine for fuel-efficient route optimization.
- To integrate all four subsystems within the unified SISURAKSHA cyber-physical ecosystem, demonstrating interoperability, modularity, and collective improvement over existing school transportation safety platforms.

CHAPTER 2: METHODOLOGY

2.1 Research Methodology Framework

The SISURAKSHA group project adopts the Design Science Research Methodology (DSRM) as the overarching research framework for the design, development, implementation, and evaluation of the proposed smart school transportation ecosystem. DSRM is a widely recognized research methodology within information systems and computing disciplines that focuses on the systematic creation and assessment of innovative technological artefacts intended to solve real-world problems through scientific and engineering principles. The methodology is particularly suitable for cyber-physical systems, IoT-based infrastructures, intelligent transportation platforms, and applied software engineering research where practical problem-solving and technological innovation are central research objectives.

The selection of DSRM for the SISURAKSHA project was motivated by the multidisciplinary nature of the identified transportation safety challenges and the requirement for iterative artefact refinement through continuous evaluation and validation. Unlike purely theoretical research methodologies, DSRM emphasizes the construction of functional technological solutions capable of addressing measurable operational deficiencies within real environments. The methodology supports the integration of hardware systems, software platforms, machine learning models, blockchain infrastructures, embedded computing architectures, and intelligent routing mechanisms into a unified research framework.

The SISURAKSHA ecosystem was developed following the core phases of Design Science Research, namely problem identification, objective definition, artefact design and development, demonstration, evaluation, and communication of research outcomes. Initially, a comprehensive analysis of Sri Lankan school transportation systems was conducted to identify critical operational weaknesses related to attendance verification, emergency management, passenger safety monitoring, route inefficiency, and

transportation accountability [1], [3], [4], [6], [31]. This phase involved reviewing accident reports, existing transportation practices, prior academic literature, and regulatory safety guidelines to establish the scope of the research problem.

Following problem identification, specific research objectives were formulated for each subsystem based on the identified gaps within existing transportation technologies. These objectives focused on developing secure attendance verification mechanisms, improving accident detection responsiveness, enhancing passenger safety monitoring, enabling movement-aware risk classification, implementing privacy-compliant biometric verification, and optimizing transportation routes dynamically through real-time attendance integration.

The artefact design and development phase involved the implementation of four interoperable but independently deployable subsystems forming the overall SISURAKSHA platform architecture. Each subsystem was developed by an individual research member while maintaining interoperability through shared APIs, communication protocols, and synchronized operational workflows. The development process incorporated IoT embedded systems, ESP32 microcontrollers, RFID modules, Edge-AI computing devices, blockchain verification services, machine learning frameworks, and full-stack web application technologies. Agile-inspired iterative prototyping techniques were employed throughout development to continuously refine subsystem performance based on testing and evaluation outcomes.

The demonstration phase involved the deployment and operational simulation of each subsystem within controlled experimental environments designed to emulate real-world school transportation scenarios. Functional validation included RFID attendance verification, blockchain hash anchoring, route proximity analysis, accident detection triggering, computer vision-based safety monitoring, footboard occupancy classification, and Edge-AI facial recognition inference. Experimental demonstrations were designed to evaluate both subsystem independence and overall ecosystem interoperability.

The evaluation phase focused on measuring system effectiveness using quantitative and qualitative performance metrics. Metrics included attendance synchronization reliability, blockchain integrity validation accuracy, accident detection responsiveness, false-positive suppression performance, machine learning inference latency, biometric recognition accuracy, route optimization efficiency, and offline operational resilience. Comparative evaluations against existing transportation practices were conducted to determine the practical contribution of the proposed ecosystem [25]-[29], [37].

Finally, the communication phase involved documenting the research contributions, implementation methodologies, system architectures, mathematical formulations, and experimental findings within this dissertation. The research outcomes collectively contribute toward the advancement of intelligent transportation safety systems within the Sri Lankan educational transportation context by demonstrating the feasibility of integrating blockchain technologies, IoT infrastructures, Edge-AI systems, and intelligent routing mechanisms into a unified child safety ecosystem.

Team Member	Subsystem / Component	Key Contributions
M. H. S. Akther (IT22590312)	Blockchain-Based RFID Attendance and Segmented Polyline Analysis	IoT Smart Attendance Device, hybrid blockchain architecture, SHA-256 hash chaining, segmented polyline algorithm, three-layer blockchain validation framework
T. A. P. K. Shameera (IT22606624)	Smart Bus Crash Detection and Safety Window Monitoring	Multi-sensor accident and hazard detection, 60-second driver verification mechanism, YOLOv8-based window safety monitoring, GSM emergency alert dispatch
P. R. I. S. R. Pinto (IT22610102)	ML and IoT Footboard Safety and Driver Monitoring	IR sensor footboard occupancy detection, Hall effect pulse-based bus movement detection, three-state risk classification, LM2596 power supply design

A. M. D. C. Kumara (IT22600134)	Smart Access and Emergency Management with Edge-AI Facial Recognition	Fine-tuned MobileFaceNet, MTCNN facial detection pipeline, AES-256 encrypted biometric template storage, CA-AEES dual-verification interlock, Attendance- Driven Dynamic Routing engine
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Table 2.1: SISURAKSHA component to team member mapping

2.2 Subsystem 1: Blockchain Based RFID Attendance and Route Matching

2.2.1 System Architecture

The blockchain-based attendance and route matching subsystem is designed as a robust three-tier distributed architecture comprising an IoT hardware layer, a backend service layer, and a front-end application layer. The architecture was specifically designed to provide tamper-resistant attendance verification, offline-first operational continuity, intelligent route assignment, and scalable synchronization capabilities suitable for real-world school transportation environments.

The IoT hardware layer is deployed directly within school buses and consists primarily of an ESP32 microcontroller integrated with an MFRC522 RFID reader module, MicroSD storage module, and wireless communication interface. Passive RFID cards assigned uniquely to students are used as identity tokens for attendance verification. When a student boards or exits the vehicle, the RFID reader captures the student identifier and forwards the attendance event to the ESP32 controller for preprocessing and temporary buffering. The ESP32 acts as the primary edge-processing unit responsible for timestamp generation, offline persistence management, duplicate validation, and deferred synchronization logic.

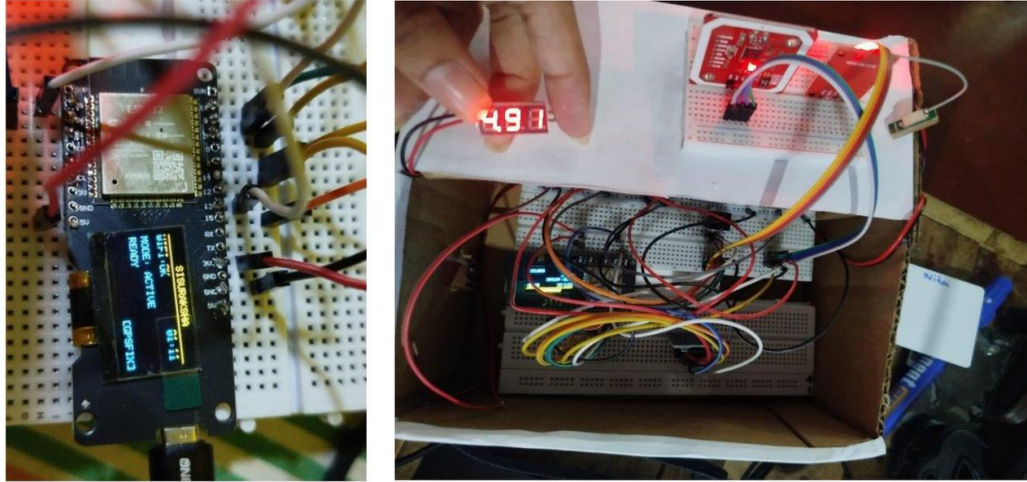


Figure 2.1: IoT Smart Attendance Device - ESP32 with PN532 and GPS Modules

A key architectural contribution of this subsystem is the implementation of an offline-first synchronization mechanism designed to address intermittent network connectivity commonly experienced within mobile transportation environments. Instead of relying exclusively on continuous internet connectivity, attendance records are first serialized locally and stored within a MicroSD-based persistence layer whenever synchronization failures occur. Once network connectivity becomes available, buffered attendance records are automatically synchronized with the backend infrastructure in chronological order. This deferred synchronization architecture significantly improves operational resilience and prevents data loss during temporary outages [10], [11], [30].

The backend service layer is implemented using a Node.js and Express.js-based RESTful API architecture integrated with MongoDB and PostgreSQL databases. MongoDB is utilized for blockchain-style attendance block persistence due to its flexible document-oriented structure, while PostgreSQL manages relational operational data such as student records, driver information, and route metadata. Attendance events received from IoT devices are grouped into logical blockchain blocks where each block

contains attendance payloads, timestamps, previous-block hashes, and SHA-256-generated cryptographic hashes to ensure tamper evidence.

To further strengthen integrity verification, the subsystem incorporates a hybrid blockchain anchoring mechanism integrated with the Ethereum blockchain network. Instead of storing full attendance payloads on-chain, only cryptographic block hashes are anchored to Ethereum using a lightweight smart contract architecture. This hybrid approach significantly reduces blockchain transaction costs while preserving public verifiability and immutable integrity validation [27]-[29]. Ethereum anchoring operations are executed asynchronously through a fire-and-forget architecture to avoid delaying IoT synchronization workflows during blockchain mining operations. Upon successful mining confirmation, the associated transaction hash is appended to the corresponding MongoDB attendance block record, thereby establishing a permanent audit trail [25], [28].

The smart contract architecture additionally emits immutable blockchain events whenever new attendance hashes are anchored. These events contain blockchain timestamps, block indices, and hash references that collectively provide a transparent and auditable historical record of attendance verification operations. Since Ethereum event logs are publicly accessible and immutable, they serve as an additional accountability layer supporting independent integrity validation and forensic verification capabilities.

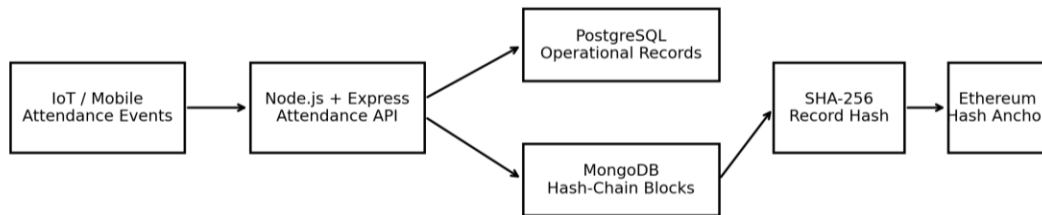


Figure 2.2: Hybrid Blockchain Architecture – Three Layer Data Flow

Another major architectural component within this subsystem is the segmented polyline analysis engine developed for intelligent driver-student matchmaking. Traditional transportation assignment systems commonly rely on Euclidean straight-line distance calculations that fail to reflect real-world road conditions [31]-[34]. To address this limitation, the subsystem integrates Open Source Routing Machine (OSRM)-generated road-network polylines for accurate spatial reasoning and transportation assignment.

When drivers register transportation routes, the backend system invokes the OSRM routing engine to generate realistic road-following route geometries represented as coordinate polylines. These polylines are stored within PostgreSQL and subsequently utilized during driver-student assignment operations. The system decomposes each route polyline into consecutive line segments and computes the minimum perpendicular point-to-segment distance between the student's home location and each segment of the driver route. The minimum computed distance value is then used to rank drivers according to real-world road-network proximity rather than simplistic geometric approximation.

The segmented polyline analysis approach models a driver's route as a collection of connected road segments generated from real road-network data rather than treating the route as a simple straight line between two locations. The system evaluates how close a student's home location is to the actual path travelled by the bus by checking the distance between the student and every road segment within the driver's route.

For each route segment, the system identifies the nearest point on the road path to the student's location and calculates the distance between them. After analyzing all segments of the route, the smallest distance value is selected as the final proximity score for that driver. This process determines how close the student actually lives to the bus route in practical road-network conditions.

Unlike traditional straight-line distance calculations, this method considers the real structure of roads, intersections, turns, and accessible driving paths. As a result, it avoids inaccurate assignments caused by obstacles such as rivers, railway crossings, restricted

roads, or disconnected streets that may appear close geographically but are operationally far apart.

Once all driver routes are evaluated, the system ranks drivers according to the calculated proximity scores. Drivers whose routes pass closest to the student receive higher priority during assignment. This segmented polyline methodology improves driver-student matchmaking accuracy, reduces unnecessary route deviations, minimizes fuel consumption, and enhances overall transportation efficiency compared to conventional distance estimation techniques [31]-[34].

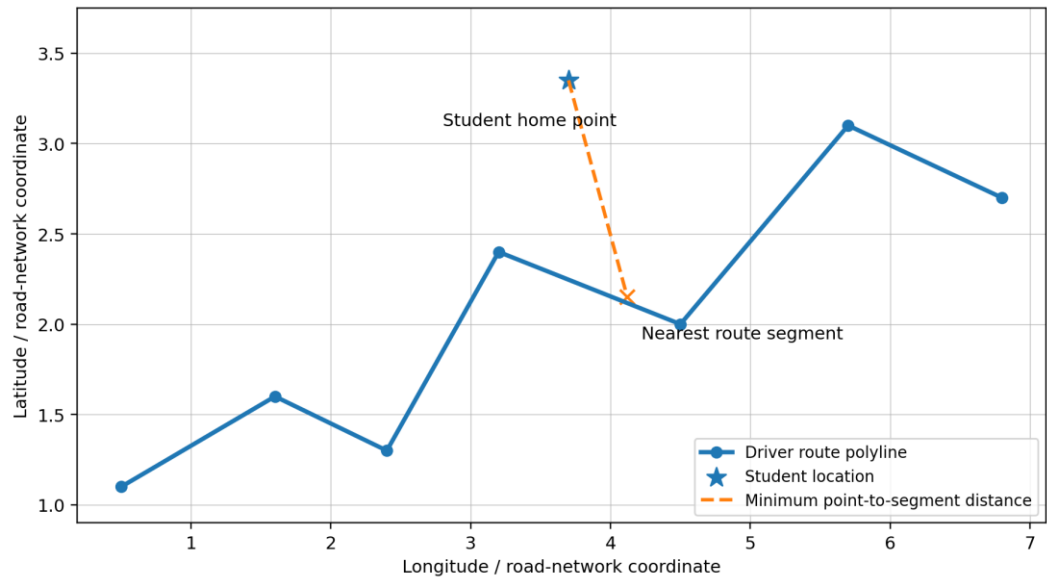


Figure 2.3: Segmented Polyline Analysis Algorithm - Driver-Student Matching

The front-end application layer is implemented using React.js based web interfaces providing role-based operational dashboards for administrators, parents, and transportation operators. Real-time attendance visibility, blockchain verification status, synchronization monitoring, route management, and transportation analytics are visualized through responsive dashboards connected to backend APIs. This multi-layer architecture collectively enables secure, scalable, and verifiable attendance management within the SISURAKSHA ecosystem [25], [26], [30].

Component	Model / Specification	Interface	Purpose
Microcontroller	ESP32 NodeMCU (240 MHz, dual-core, 4 MB flash)	GPIO, SPI, I2C, UART	Main processing, Wi-Fi, and logic control
RFID/NFC module	PN532 v1.6 (13.56 MHz, ISO 14443A/B)	SPI (HSPI)	RFID card UID reading
GPS module	NEO-6M / NEO-8M (1-5 m accuracy)	UART (Serial2)	Real-time bus location capture
MicroSD module	SPI MicroSD adapter (FAT32, up to 32 GB)	SPI (VSPI)	Offline attendance log storage
OLED display	0.96 inch SSD1306 OLED (128 x 64 pixels)	I2C	Status display and feedback
Passive buzzer	5V passive buzzer module	GPIO PWM	Audio scan confirmation
Power supply	5V via USB / DC input	-	Device power

Table 2.2: IoT Smart Attendance Device Hardware Components

2.3 Subsystem 2: Crash Detection and Window Safety Monitoring

2.3.1 Accident Detection System Design

The crash detection and safety monitoring subsystem employs an ESP32 microcontroller as the central processing unit integrated with a comprehensive multi-sensor array specifically selected to detect accident conditions, environmental hazards, and passenger safety threats in real time. The subsystem was designed to provide rapid emergency detection, false alert reduction, and automated emergency escalation within school transportation environments.

The hardware architecture integrates multiple sensing modules including accelerometers, vibration sensors, tilt sensors, gas sensors, smoke sensors, water detection sensors, and pressure monitoring units. Each sensor contributes to a sensor-fusion decision-making framework capable of identifying various hazardous operational conditions including

vehicle rollovers, severe collisions, abnormal vibration events, fire outbreaks, smoke accumulation, gas leakage, flooding, and environmental anomalies [16], [17], [24].

The ESP32 microcontroller continuously collects sensor readings through high-frequency polling operations and evaluates sensor thresholds against predefined hazard classification conditions. Rather than relying on isolated sensor triggers, the subsystem adopts a multi-sensor fusion methodology in which emergency events are validated through correlated sensor activation patterns. This significantly reduces false-positive activations caused by potholes, abrupt braking, or uneven terrain conditions commonly encountered during transportation operations.

One of the key innovations of this subsystem is the implementation of a 60-second driver verification mechanism designed to suppress accidental emergency alerts. Upon detection of a potential crash or hazardous event, the system initiates an emergency countdown timer while simultaneously notifying the driver through onboard audiovisual alert mechanisms. During this verification interval, the driver may manually cancel the alert if the detected event does not represent a genuine emergency condition. If no cancellation occurs within the predefined interval, the system automatically escalates the event and transmits emergency notifications to predefined stakeholders including parents, administrators, and emergency responders.

The subsystem additionally incorporates a YOLOv8-based computer vision safety monitoring module focused on passenger window behavior analysis. An onboard camera continuously captures visual frames within the passenger cabin while the YOLOv8 object detection model performs real-time behavioral inference directly on edge hardware. The model was trained to identify unsafe window-related behavior patterns including excessive leaning, partial body exposure, and dangerous passenger positioning near windows.

Detected unsafe behaviors trigger immediate warning notifications to transportation staff while simultaneously logging safety incidents within the centralized monitoring

platform. The low-latency inference capabilities of YOLOv8 make the architecture suitable for real-time transportation environments where immediate hazard recognition is essential.

Collectively, this subsystem combines sensor fusion, embedded IoT processing, and real-time computer vision analysis to provide a proactive transportation safety architecture capable of improving emergency responsiveness, reducing accident severity, and enhancing passenger safety monitoring within school transportation ecosystems [4], [19]-[21], [24].

Component / Sensor	Purpose in the System
ESP32	Main microcontroller for sensor processing and control
MPU9250	Detects acceleration, tilt, orientation and rollover conditions
801S vibration sensor	Detects abnormal vibration and impact
Piezoelectric sensor	Detects shock or collision-like impact
Force sensitive resistor	Detects high pressure or force conditions
NEO-6M GPS module	Retrieves bus location during confirmed accidents
SIM800L GSM module	Sends emergency SMS alerts
Push button	Allows driver to cancel false accident alerts
DFPlayer Mini / speaker	Provides audio warnings
MQ2 gas sensor	Detects smoke or gas-related hazards
Fire sensor	Detects fire or flame-related hazards
Raindrop sensor	Detects water or flood-related exposure
LM2596 buck converter	Provides stable voltage regulation for modules
BMP280	Supports environmental monitoring

Table 2.3: Accident and Emergency Sensor Integration Summary

2.4 Subsystem 3: Footboard Safety and Driver Monitoring

2.4.1 Footboard Safety Architecture

The footboard safety monitoring subsystem focuses specifically on the school bus entrance and footboard region, which constitutes one of the highest-risk operational zones within school transportation environments. Boarding and alighting accidents involving moving vehicles continue to represent major safety concerns due to inadequate monitoring and the absence of movement-aware occupancy classification systems.

The subsystem architecture combines infrared proximity sensing, Hall effect pulse-based movement detection, embedded edge processing, and real-time alert generation to provide contextual footboard safety monitoring. Infrared sensors are strategically positioned near the entrance and footboard area to continuously monitor passenger occupancy conditions. These sensors detect the presence of passengers within predefined risk zones and transmit occupancy information to the embedded processing unit.

IR Sensor Placement on School Bus Footboard

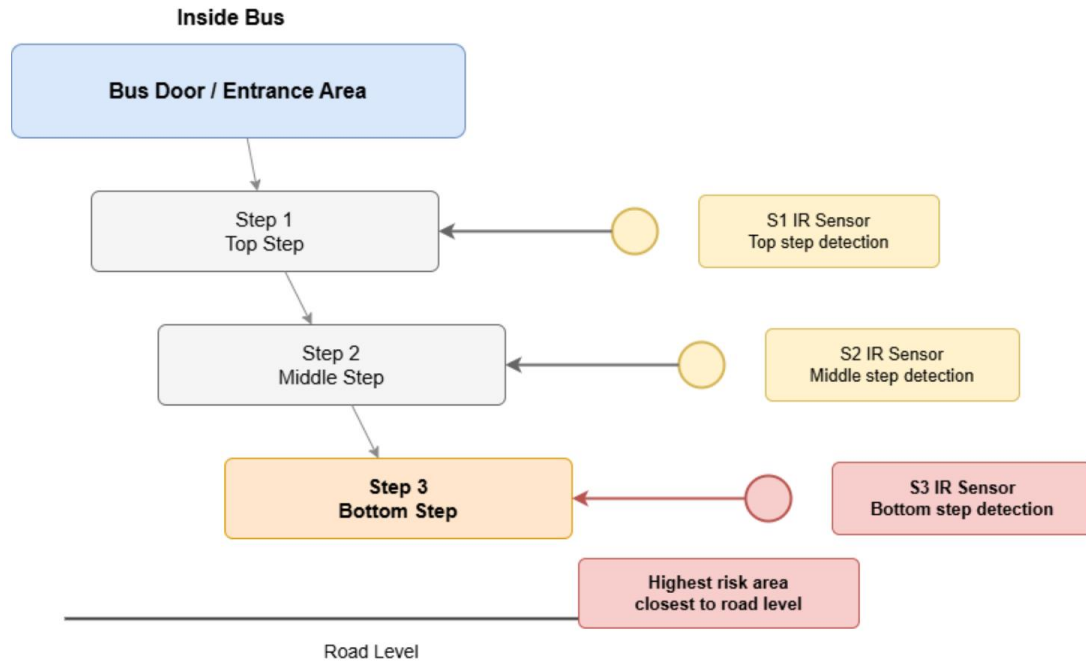


Figure 2.4: Footboard Safety Module - IR Sensor Placement on Bus Entrance

A major limitation of existing transportation safety systems is the dependence on GPS-based movement estimation for determining whether a vehicle is stationary or in motion. GPS-based approaches frequently suffer from signal instability, delayed updates, and reduced accuracy in obstructed operational environments [18], [23]. To overcome these limitations, the subsystem integrates a Hall effect sensor attached to the vehicle's rotational movement mechanism for direct pulse-based motion analysis.

The Hall effect sensor continuously monitors rotational magnetic field changes associated with wheel or propeller shaft movement. Pulse frequencies generated by rotational activity are analyzed by the microcontroller to determine whether the vehicle is stationary, moving slowly, or operating under normal transit conditions. This movement-aware approach provides significantly more reliable motion classification than GPS-dependent alternatives.

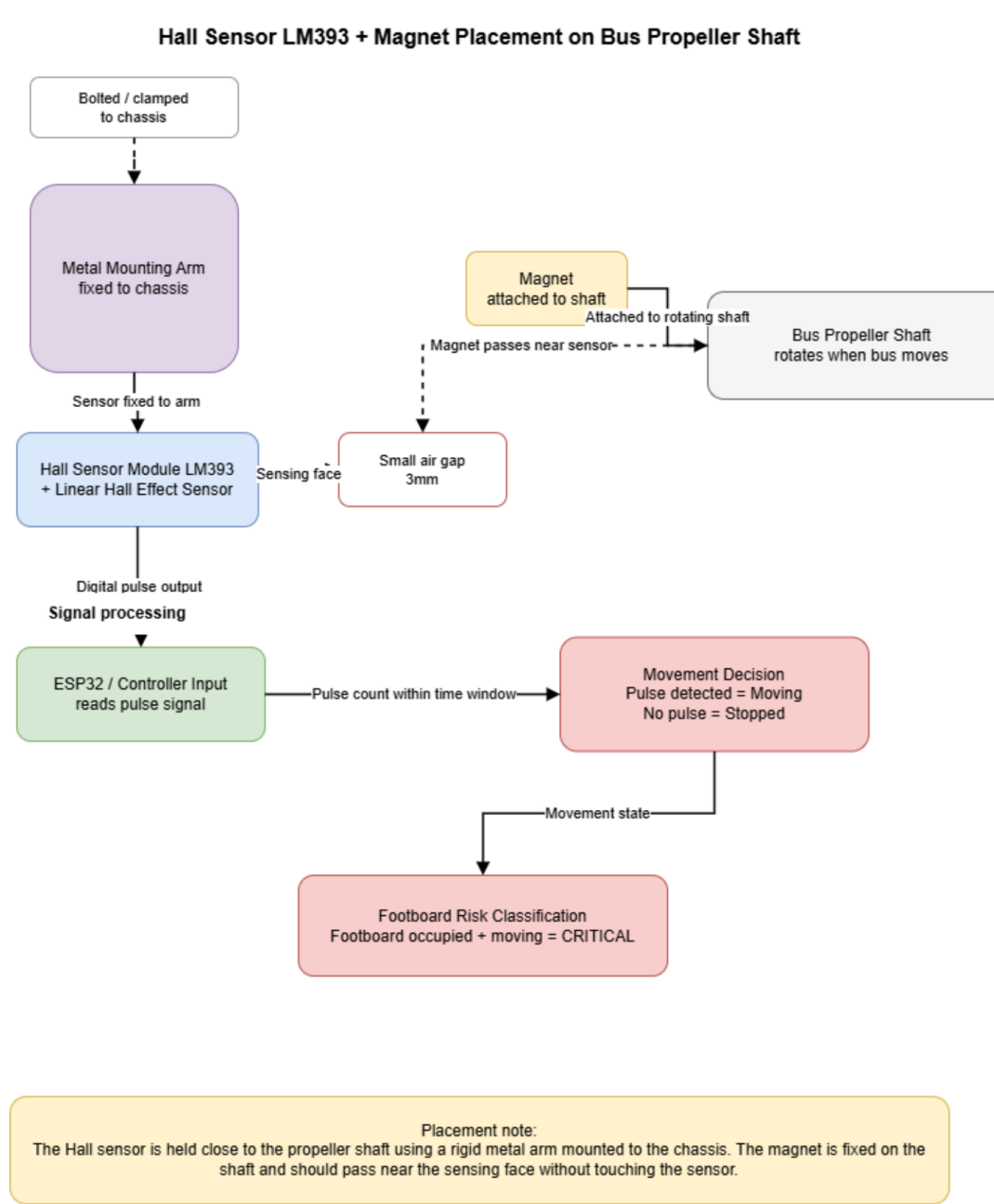


Figure 2.5: Hall Sensor Movement Detection - Propeller Shaft Mounting

The subsystem implements a three-state contextual safety classification model consisting of SAFE, WARNING, and CRITICAL operational states. When passenger occupancy is detected while the vehicle remains stationary, the system classifies the condition as SAFE. If occupancy persists during low-speed movement, the state transitions into

WARNING mode. When dangerous occupancy conditions are detected during active vehicle movement, the subsystem enters CRITICAL mode and immediately triggers audiovisual alerts and notification escalation workflows.

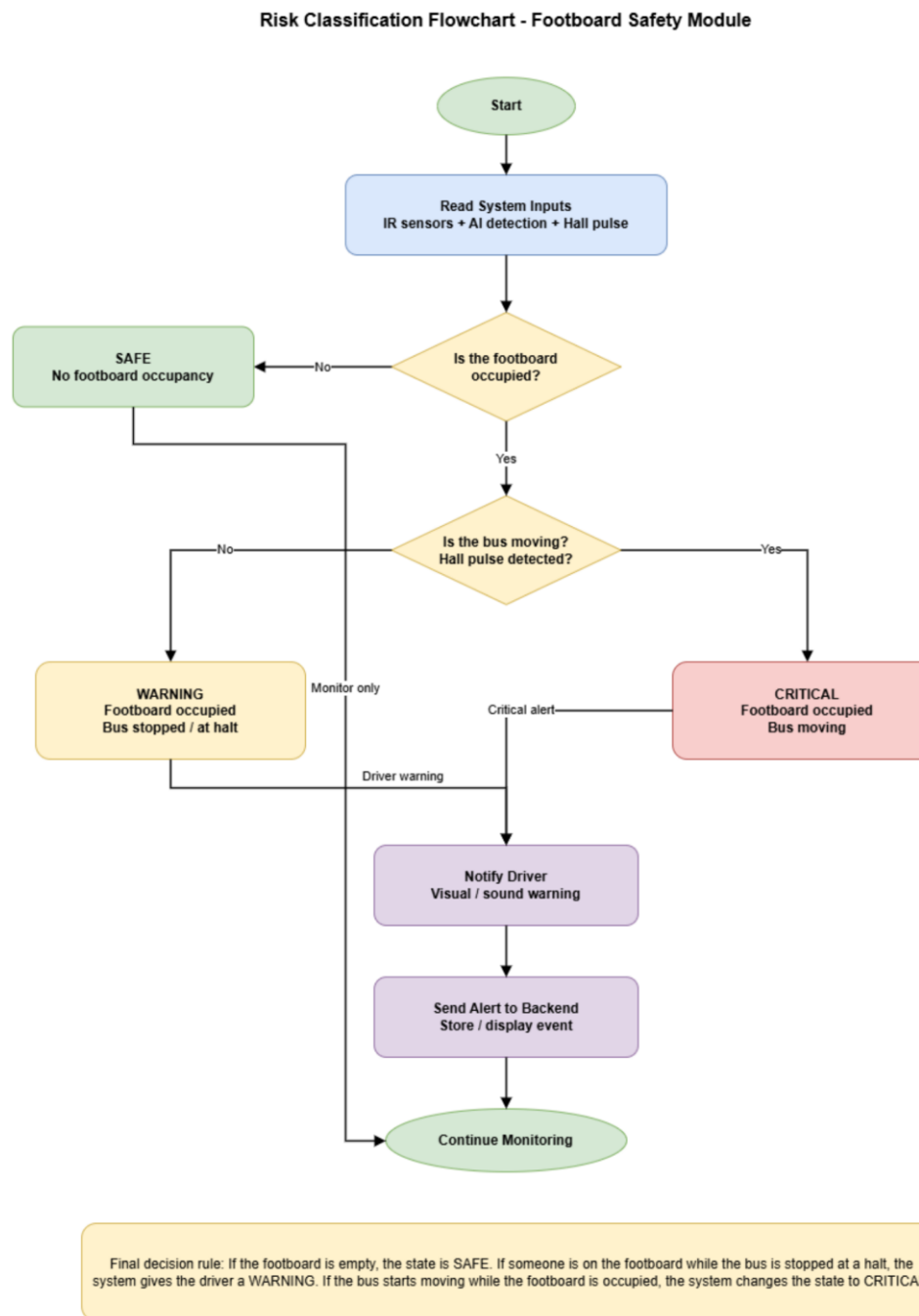


Figure 2.6: Risk Classification State Machine

Real-time alerts are communicated through onboard warning systems while event data is simultaneously transmitted to the centralized monitoring platform for administrative visibility and incident logging. The architecture also supports future integration with automated vehicle immobilization systems capable of preventing unsafe bus movement under hazardous occupancy conditions.

The subsystem significantly improves situational awareness around one of the most accident-prone areas within school transportation operations by combining reliable movement classification with contextual occupancy monitoring and intelligent risk evaluation [22], [23].

Footboard Condition	Movement State	System State
No occupancy	Stopped or moving	SAFE
Occupied	Stopped / at halt	WARNING to driver
Occupied	Moving / Hall pulse detected	CRITICAL alert
Uncertain sensor or AI reading	Any	WARNING or manual check

Table 2.4: Footboard Safety Risk Classification Logic

2.5 Subsystem 4: Edge AI Facial Recognition and Emergency Management

2.5.1 Edge AI Biometric Architecture

The smart access and emergency management subsystem is built upon a fully decentralized Edge-AI biometric architecture designed specifically for privacy-preserving student verification and strict compliance with Sri Lanka’s Personal Data Protection Act (PDPA) No. 9 of 2022 [38]-[40]. The subsystem integrates biometric identity verification, intelligent emergency management, localized AI inference, and attendance-driven route optimization into a unified edge-computing framework.

The biometric architecture utilizes MobileFaceNet as the primary facial recognition model due to its lightweight computational requirements, low inference latency, and suitability for deployment on resource-constrained edge devices. Face detection and preprocessing operations are performed using Multi-task Cascaded Convolutional Networks (MTCNN), enabling reliable facial localization under varying environmental conditions including changes in illumination, passenger positioning, and camera angle variations.

Unlike conventional cloud-dependent biometric systems, the SISURAKSHA subsystem performs all facial recognition inference operations locally on edge hardware platforms such as the NVIDIA Jetson Nano. This decentralized architecture minimizes network dependency, reduces bandwidth consumption, improves inference responsiveness, and significantly enhances privacy protection by preventing unnecessary biometric data transmission to external cloud servers [38]-[42].

Biometric templates and facial embeddings are encrypted and stored locally within protected edge storage environments to minimize exposure risks associated with centralized cloud persistence. Localized inference also improves compliance with PDPA regulatory requirements governing the processing of sensitive biometric information involving minors.

During boarding operations, onboard cameras capture student facial images and forward them through the MTCNN preprocessing pipeline before generating feature embeddings using MobileFaceNet. Generated embeddings are then compared against locally stored encrypted biometric templates using cosine similarity-based matching techniques. Successfully matched students are authenticated automatically, and attendance events are forwarded to the centralized transportation platform.

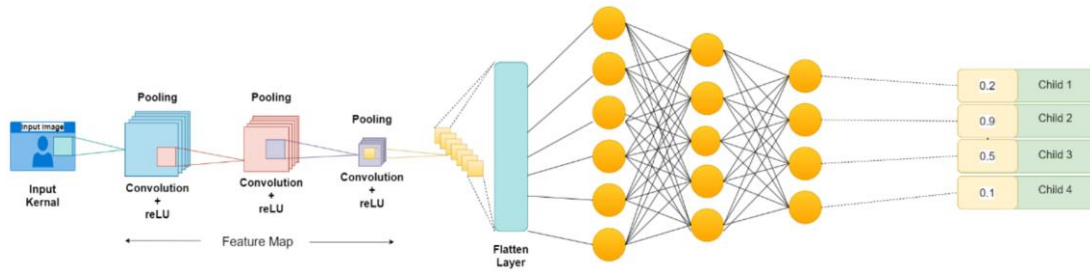


Figure 2.7: Edge-Ai biometric pipeline - Mtcnn and Mobilefacenet architecture

In addition to biometric verification, the subsystem incorporates a Context-Aware Automated Emergency Egress System (CA-AEES) designed to improve emergency evacuation responsiveness during critical transportation incidents. The emergency management framework evaluates environmental sensor inputs, accident conditions, and operational states before triggering automated emergency workflows including door control assistance, emergency alert escalation, and evacuation coordination support.

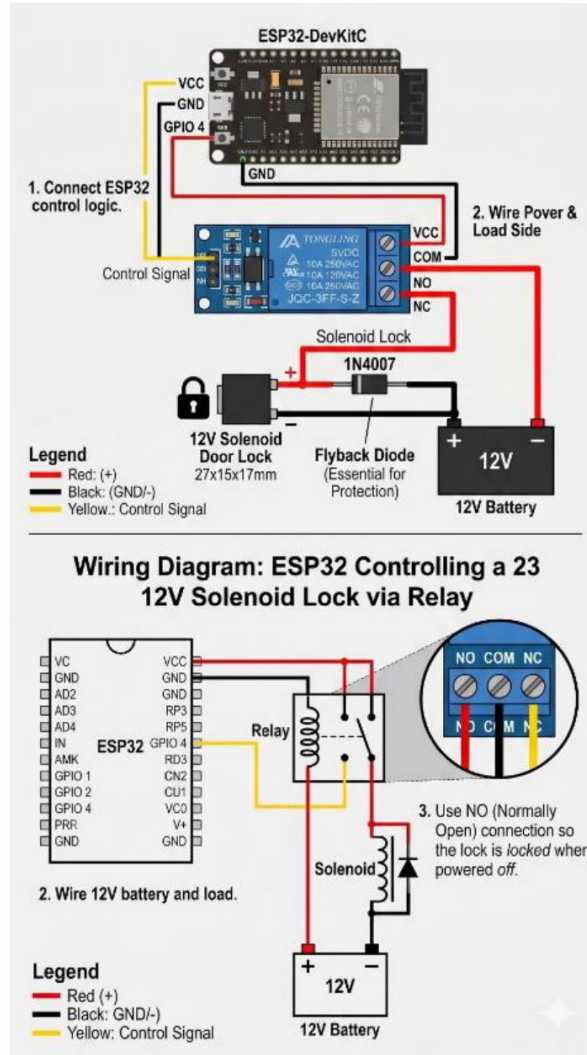


Figure 2.8: CA-AEES dual verification interlock - sensor fusion flowchart

Another major contribution of this subsystem is the implementation of an Attendance-Driven Dynamic Routing engine. Traditional school transportation systems generally rely on static route configurations that continue visiting student pickup locations regardless of daily attendance status [6], [31], [43], [44]. This results in unnecessary fuel consumption, longer travel durations, and operational inefficiency.

To address this issue, the dynamic routing engine continuously evaluates verified attendance data and automatically prunes unnecessary route nodes associated with

absent students. Graph-based optimization algorithms are then applied to generate more efficient transportation paths based on active attendance conditions. This attendance-aware routing approach reduces fuel wastage, improves transportation efficiency, and minimizes ghost-stop operational overhead within school transportation ecosystems.

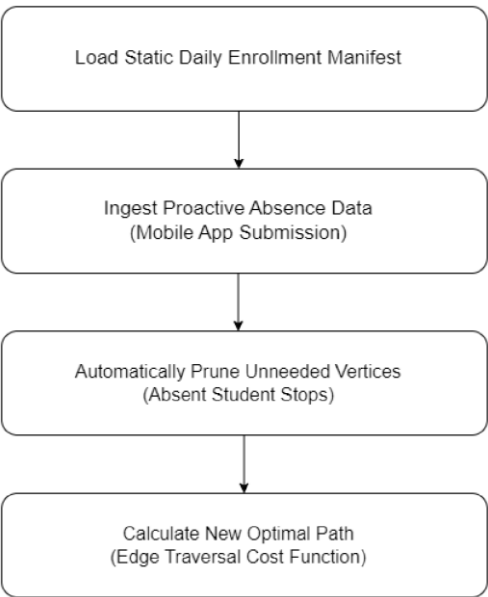


Figure 2.9: attendance driven dynamic routing engine - graph pruning diagram

Through the integration of Edge-AI biometric verification, decentralized privacy-preserving inference, intelligent emergency management, and attendance-driven transportation optimization, this subsystem contributes significantly toward the realization of a proactive and intelligent smart school transportation ecosystem [6], [38], [43], [44].

CNN Architecture	Finalized Accuracy	Average Inference Latency
VGG-Face (baseline)	88.45%	850 ms
ResNet50	92.10%	540 ms
MobileNetV2	94.33%	210 ms

InceptionResNetV1 (standard FaceNet)	98.85%	480 ms
MobileFaceNet	99.12%	120 ms

Table 2.5: CNN Model Comparison for Edge-AI Biometric Classification

2.6 Integrated System Architecture

The four subsystems of SISURAKSHA are designed to operate as modular but interoperable components within a shared system architecture. The common backend infrastructure, built on Node.js and Express.js with PostgreSQL and MongoDB persistence, provides a unified data layer accessible to all subsystems through defined RESTful API contracts.

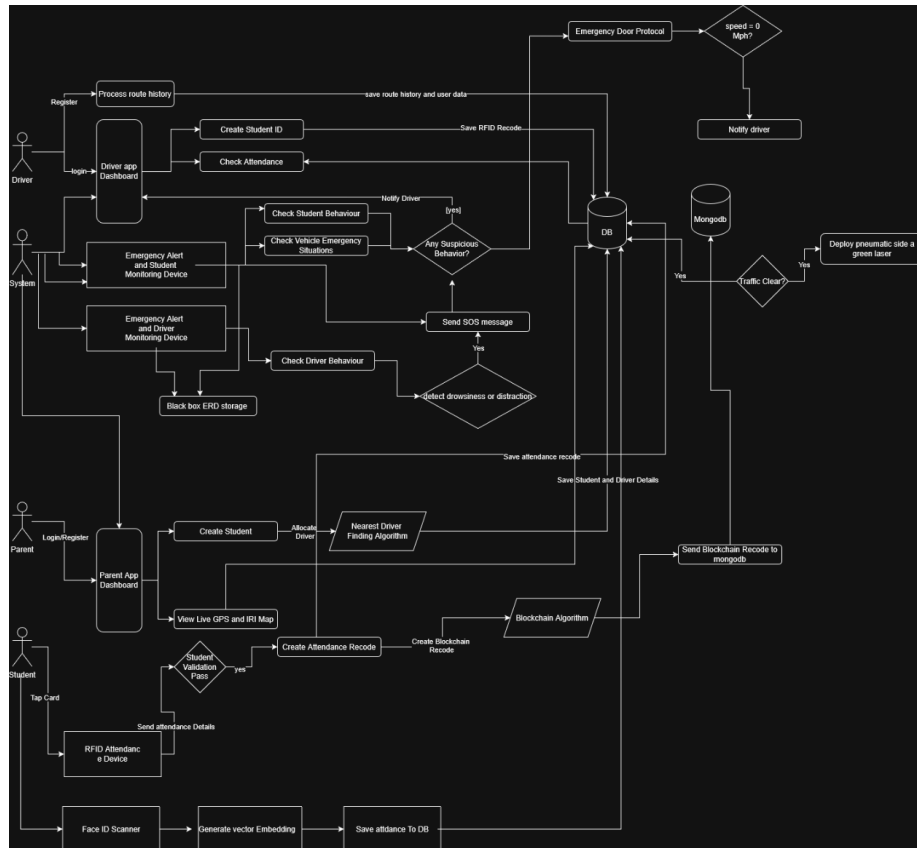


Figure 2.10: SISURAKSHA ecosystem architecture overview

2.7 Commercialization Aspects

The SISURAKSHA platform presents a compelling commercial proposition addressing a clearly documented safety, accountability, and operational efficiency requirement within Sri Lanka's school transportation ecosystem. Increasing parental concern regarding student safety, growing institutional pressure for operational transparency, and the rising frequency of transportation-related incidents collectively create a strong market demand for intelligent school transportation technologies. Unlike conventional GPS-only school tracking systems currently available within the local market, SISURAKSHA offers a comprehensive cyber-physical ecosystem integrating blockchain-secured attendance verification, Edge-AI biometric authentication, real-time accident detection, passenger safety monitoring, and attendance-driven route optimization into a unified operational platform.

The proposed commercialization model follows a hybrid hardware-software deployment strategy combining one-time infrastructure installation revenue with recurring subscription-based software and monitoring services. Under this model, transportation providers, schools, and private fleet operators would initially incur a capital expenditure associated with the purchase and installation of SISURAKSHA-compatible hardware components. These include IoT edge-computing devices, RFID modules, biometric access units, ESP32 sensor controllers, Hall effect movement sensors, emergency safety modules, onboard cameras, and automated interlock mechanisms integrated into existing school bus fleets.

The hardware deployment phase represents the primary onboarding revenue stream and includes installation, calibration, device configuration, route registration, and subsystem integration services. Because the SISURAKSHA architecture is modular, institutions may adopt individual subsystems independently or deploy the complete integrated ecosystem based on operational requirements and budget constraints. This modular commercialization approach improves scalability and increases market accessibility for

small and medium-scale transportation operators who may not initially possess the financial capacity for full-system deployment.

In addition to hardware deployment revenue, SISURAKSHA supports a Software-as-a-Service (SaaS) subscription model for ongoing platform access, cloud synchronization, analytics dashboards, blockchain verification services, and centralized monitoring operations. Subscription packages may be tiered according to fleet size, number of registered students, frequency of blockchain transactions, AI monitoring capabilities, and emergency response integration requirements. Such recurring subscription structures create sustainable long-term revenue generation while ensuring continuous software maintenance, security updates, and infrastructure support.

The blockchain-based attendance subsystem introduces a unique commercial differentiator within the educational transportation technology market. By implementing a hybrid blockchain architecture that anchors only cryptographic attendance hashes onto Ethereum while maintaining operational data within off-chain MongoDB infrastructure, SISURAKSHA significantly reduces blockchain operational costs compared to fully decentralized systems [27]-[29]. This cost-efficient integrity verification model improves commercial viability by enabling large-scale deployment without excessive transaction expenditure, thereby making the platform economically practical for local school transportation operators.

Another major commercialization advantage arises from the platform's attendance-driven dynamic routing capabilities. Conventional school transportation systems frequently waste fuel and operational resources through static routing strategies that continue servicing absent students, commonly referred to as ghost-stop inefficiency [6], [31], [43], [44]. The SISURAKSHA routing engine reduces unnecessary travel distance through graph-based route pruning and attendance-aware optimization, thereby lowering fuel consumption, vehicle wear, and transportation overhead. These measurable

operational savings create strong financial incentives for private bus owners and school administrators to adopt the platform.

The Edge-AI biometric subsystem also contributes substantial commercial value by enabling privacy-preserving student verification without continuous cloud dependency. Since biometric inference is performed locally on edge hardware, bandwidth usage and cloud-processing costs are significantly reduced while simultaneously improving compliance with Sri Lanka's Personal Data Protection Act (PDPA) No. 9 of 2022 [38]-[40]. This decentralized architecture improves scalability and positions SISURAKSHA as a privacy-conscious transportation technology solution suitable for regulatory-sensitive educational environments.

Beyond direct institutional deployment, SISURAKSHA also possesses commercialization potential through strategic partnerships with insurance providers, educational authorities, transportation regulators, and smart-city infrastructure initiatives. Insurance companies may provide premium reductions for fleets equipped with verified accident detection and safety-monitoring capabilities, while educational institutions may utilize the platform as part of broader student safety assurance programs. Similarly, government agencies and provincial transportation authorities may adopt such systems within future smart transportation modernization initiatives aimed at improving national child transportation safety standards.

The platform's modular architecture additionally enables future international scalability and adaptation beyond Sri Lanka. Many developing countries experience similar school transportation challenges including weak attendance accountability, limited safety automation, manual emergency management, and inefficient routing practices. Consequently, SISURAKSHA possesses long-term export and regional commercialization potential within broader South Asian and developing-market educational transportation sectors.

From a sustainability perspective, the recurring subscription model combined with scalable subsystem deployment creates opportunities for long-term operational continuity and incremental feature expansion. Future monetization opportunities may include predictive maintenance analytics, AI-driven driver behavior assessment, fleet performance optimization, insurance risk scoring, and smart-city transportation integration services. These expansion pathways position SISURAKSHA not merely as an academic prototype, but as a commercially viable intelligent transportation platform with strong real-world deployment potential across educational transportation ecosystems.

CHAPTER 3: RESULTS AND DISCUSSION

3.1 Subsystem 1 Results: Blockchain-Based RFID Attendance and Route Matching

The Blockchain-Based RFID Attendance and Route Matching subsystem was evaluated through a combination of hardware validation tests, network synchronization experiments, blockchain verification analysis, and routing intelligence simulations. The experimental evaluation was conducted to measure the operational reliability, responsiveness, offline resilience, and scalability of the proposed attendance ecosystem under realistic transportation conditions.

The IoT Smart Attendance Device was evaluated across 200 simulated RFID attendance scan events using 13.56 MHz MIFARE Classic 1K RFID cards. The PN532 RFID reader module successfully detected and authenticated all presented cards within the expected operational read range of 0–4 cm, achieving a 100% read success rate consistent with manufacturer specifications and prior RFID performance studies [8], [9], [15]. No duplicate reads or corrupted attendance payloads were observed during testing, indicating stable hardware-level communication between the RFID module and the ESP32 microcontroller.

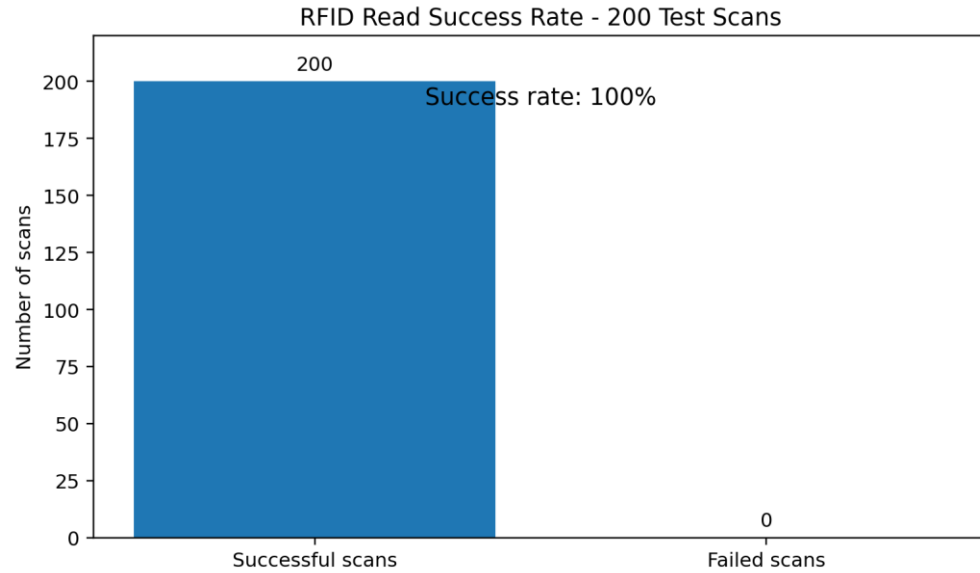


Figure 3.1: RFID read success rate

The integrated GPS module was tested under both open-sky and partially obstructed environmental conditions to evaluate route-registration and location-awareness reliability. Under unobstructed outdoor conditions, the GPS module achieved an average cold-start fix acquisition time of 45 seconds. Under partially obstructed conditions involving nearby structures and intermittent signal interference, the average fix acquisition time increased to approximately 90 seconds. While obstructed acquisition times were higher, the module still maintained acceptable operational functionality suitable for school transportation route initialization.

Offline persistence reliability was evaluated through repeated local storage and deferred synchronization tests using the MicroSD-based offline-first architecture. During simulated network outages, attendance records were successfully serialized and written locally to the FAT32-formatted MicroSD storage with an average write latency of 12 milliseconds per JSON attendance record. Buffered records were subsequently synchronized automatically once network connectivity was restored. Across 200 buffered attendance records, the deferred synchronization mechanism achieved a 99.5%

synchronization success rate, demonstrating strong resilience against temporary network instability and confirming the effectiveness of the offline-first IoT architecture.

Network synchronization performance was additionally evaluated under varying wireless signal conditions. Under stable 2.4 GHz Wi-Fi connectivity, HTTP POST attendance synchronization requests achieved an average latency of 380 milliseconds. Under weak signal conditions characterized by low wireless signal strength (1–2 signal bars), average synchronization latency increased to approximately 1,200 milliseconds. Despite this increase, all synchronization requests remained operationally acceptable within the system timeout threshold and did not trigger unintended offline fallback operations.

The blockchain integrity verification mechanism was evaluated by testing the SHA-256 hash chaining architecture across multiple attendance blocks. Experimental tampering simulations demonstrated that any modification to historical attendance payloads immediately altered subsequent cryptographic hashes, thereby invalidating blockchain continuity and exposing unauthorized data modification attempts. This confirms the effectiveness of the hybrid blockchain architecture in providing tamper-evident attendance verification without requiring full on-chain storage of attendance records [27]-[29].

Ethereum anchoring functionality was also tested using the deployed smart contract infrastructure. Blockchain transaction hashes were successfully generated and recorded for all evaluated attendance blocks. The asynchronous fire-and-forget anchoring architecture significantly improved API responsiveness by allowing IoT synchronization operations to complete without waiting for blockchain mining confirmation. This design prevented synchronization bottlenecks and improved overall operational scalability during high-frequency attendance submission scenarios.

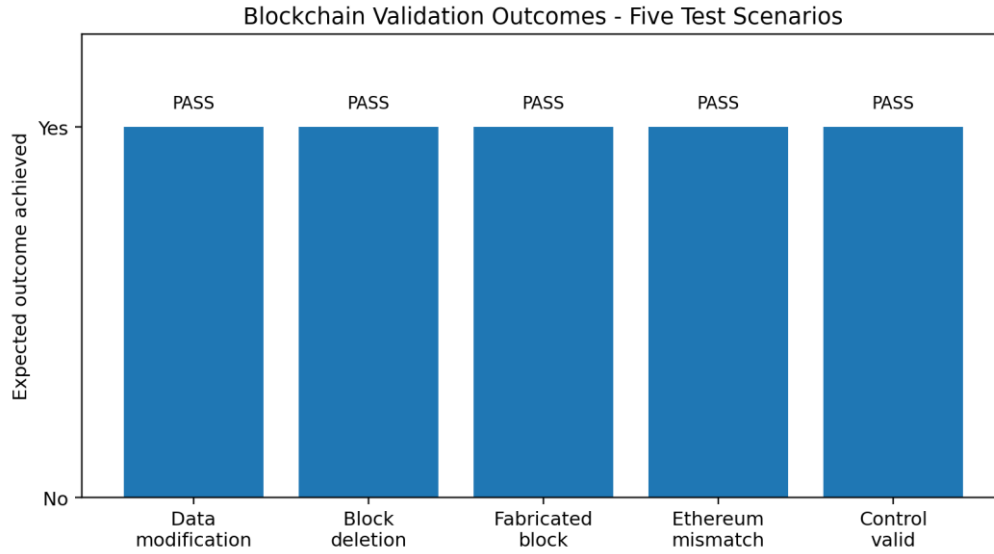


Figure 3.2: blockchain tamper detection

The segmented polyline analysis engine was evaluated using multiple simulated driver routes and student pickup coordinates generated through the Open Source Routing Machine (OSRM). Experimental comparisons between traditional straight-line distance assignment and segmented road-network polyline analysis demonstrated that the proposed methodology produced significantly more realistic driver-student assignment recommendations. By computing minimum point-to-segment distances against actual navigable road geometries, the system avoided inaccurate route assignments commonly produced by Euclidean distance approximation methods [31]-[34].

The overall subsystem evaluation confirmed that the proposed architecture successfully achieved secure attendance verification, resilient offline synchronization, low-latency IoT communication, scalable blockchain anchoring, and realistic route-based driver assignment suitable for deployment within school transportation environments.

Metric	Value	Condition
RFID read success rate	100%	200 test scans, 0-4 cm range
GPS fix time (cold start)	45 s (avg)	Open sky
GPS fix time (obstructed)	90 s (avg)	Partial obstruction

SD write latency	12 ms (avg)	FAT32, JSON record
HTTP POST latency (good signal)	380 ms (avg)	Stable 2.4 GHz Wi-Fi
HTTP POST latency (weak signal)	1,200 ms (avg)	Weak Wi-Fi, 1-2 bar
Offline buffer sync success rate	99.5%	200 buffered records batch sync

Table 3.1: IoT smart attendance device performance metrics

Tamper Scenario	Detection Result	Detected By
Direct modification of attendance record in MongoDB	detected	Data integrity check: hash recomputation mismatch
Deletion of a blockchain block document	detected	Chain integrity check: previousHash mismatch at block + 1
Insertion of fabricated block with invalid hash	detected	Data integrity check: invalid hash format and value
Ethereum txHash field modification in MongoDB	detected	Ethereum cross-validation: on-chain vs local mismatch
Unmodified control scenario (no tampering)	valid - no false positive	All three checks passed

Table 3.2: blockchain validation framework - tamper detection test outcomes

3.2 Subsystem 2 Results: Crash Detection and Window Safety

The Crash Detection and Window Safety Monitoring subsystem was evaluated through systematic experimental testing covering all major accident, hazard, and passenger safety scenarios defined during the system design phase. The evaluation focused on sensor reliability, emergency responsiveness, false-alert suppression effectiveness, and real-time computer vision inference performance.

The accident detection subsystem was tested using nine distinct operational scenarios including rollover detection, vibration anomaly sensing, severe impact detection, pressure variation analysis, fire detection, smoke and gas leakage detection, water hazard detection, driver verification and cancellation workflows, and confirmed emergency escalation with SMS notification dispatch. Each scenario was executed repeatedly under controlled simulated conditions to validate sensor stability and multi-sensor fusion reliability.

The accelerometer and tilt sensor combination successfully detected rollover and severe impact conditions with high consistency across all experimental runs. Multi-sensor fusion significantly reduced false-positive activations compared to single-sensor threshold approaches by requiring correlated sensor activation patterns before triggering emergency escalation workflows. During pothole and abrupt braking simulations, the system correctly identified non-critical operational disturbances and avoided unnecessary emergency escalation events.

The integrated smoke, gas, and fire detection sensors demonstrated reliable environmental hazard identification capabilities under controlled hazardous condition simulations. Water intrusion detection mechanisms also successfully identified simulated flooding scenarios relevant to submerged vehicle emergencies. The subsystem continuously monitored sensor thresholds in real time and initiated emergency evaluation procedures immediately upon threshold violation detection.

A major focus of subsystem evaluation involved assessing the effectiveness of the 60-second driver verification and cancellation mechanism. Experimental results demonstrated that the verification interval significantly reduced false emergency alerts generated by non-critical disturbances. Drivers were successfully able to cancel accidental emergency triggers within the verification window during controlled non-emergency simulations, thereby preventing unnecessary stakeholder notification escalation.

The emergency notification subsystem successfully dispatched SMS alerts containing vehicle identification and emergency status information following confirmed incident conditions. Alert transmission delays remained operationally acceptable under standard mobile network conditions, thereby supporting rapid emergency communication workflows.

The YOLOv8-based passenger window safety monitoring component was evaluated using multiple simulated unsafe passenger behavior scenarios including leaning outside windows, partial body exposure, and unsafe seating positions. The model successfully detected unsafe passenger behaviors in real time with low inference latency suitable for live transportation environments. Detection consistency remained stable under varying passenger positioning conditions and moderate lighting variation scenarios.

Experimental testing confirmed that the integration of multi-sensor fusion and embedded computer vision analysis substantially improved proactive hazard detection capabilities within school transportation environments. The subsystem successfully demonstrated reliable accident recognition, false-alert reduction, environmental hazard monitoring, emergency escalation automation, and passenger behavior analysis suitable for real-world deployment [16], [19], [20], [24].

Test Scenario	Expected Input / Condition	Expected Output
Simulate rollover / tilt condition	MPU9250 exceeds tilt threshold	Possible accident detected
Apply strong vibration to sensor	801S vibration threshold exceeded	Abnormal vibration detected
Apply shock to piezoelectric sensor	Piezoelectric value exceeds threshold	Shock / impact detected
Apply pressure to FSR	FSR value increases above threshold	High pressure condition detected
Trigger possible accident	Button pressed within 60	False alert cancelled

and press cancel button	seconds	
Trigger possible accident without cancellation	No button press within 60 seconds	Confirmed accident and SMS alert sent
Activate fire sensor	Fire threshold triggered	Fire hazard detected
Expose MQ2 to smoke/gas test condition	MQ2 value exceeds threshold	Smoke/gas hazard detected
Expose raindrop sensor to water	Water sensor value changes	Water hazard warning generated

Table 3.3: accident detection test cases and results

Test Scenario	Expected Input / Condition	Expected Output
Student hand inside the bus window	Safe image/frame	No warning generated
Student hand outside the window	Hand outside window detected	Audio warning activated

Table 3.4: window safety detection test cases and results

3.3 Subsystem 3 Results: Footboard Safety and Driver Monitoring

The Footboard Safety and Driver Monitoring subsystem was evaluated through a series of operational simulations designed to validate movement classification accuracy, occupancy detection reliability, and contextual safety-state transitions under varying transportation conditions.

Prototype testing confirmed the correct operation of the three-state contextual footboard safety classification logic consisting of SAFE, WARNING, and CRITICAL operational states. During stationary bus simulations where no passenger occupancy was detected

near the entrance region, the subsystem correctly maintained the SAFE operational state without generating unnecessary alerts.

When infrared sensors detected passenger occupancy while the vehicle remained stationary, the subsystem appropriately continued operating within the SAFE state because boarding and alighting operations under stationary conditions were classified as non-hazardous. This demonstrated the effectiveness of contextual occupancy evaluation rather than simplistic occupancy-only alert triggering.

The Hall effect pulse-based movement detection mechanism was evaluated across multiple vehicle movement simulations. The sensor successfully identified rotational movement conditions through magnetic pulse frequency analysis and consistently differentiated between stationary and moving operational states. Unlike GPS-based movement estimation systems, the Hall effect architecture demonstrated stable operation without dependence on satellite signal availability or network connectivity [18], [23].

During low-speed vehicle movement simulations involving detected passenger presence near the footboard, the subsystem correctly transitioned into the WARNING state and activated preliminary alert notifications. Under higher-risk movement conditions where occupancy persisted during active bus motion, the subsystem reliably entered the CRITICAL state and triggered audiovisual warning mechanisms immediately.

Experimental evaluation additionally confirmed that the Hall effect pulse-based movement classification mechanism produced highly reliable binary movement detection without false movement triggers during stationary conditions. Infrared occupancy sensors also maintained stable proximity detection performance throughout repeated experimental trials.



Figure 3.3: footboard safety prototype - safe, warning, and critical state outputs

The integrated architecture demonstrated the practical feasibility of movement-aware footboard safety monitoring through contextual risk classification and reliable movement analysis. Results confirmed that the subsystem effectively improved situational awareness around one of the highest-risk operational zones within school transportation systems.

Test ID	Test Scenario	Observed Result
PT-01	No object/person near footboard	SAFE state captured
PT-02	Object/person detected while stopped	WARNING state captured
PT-03	Footboard occupied with movement condition	CRITICAL state captured

PT-04	ESP32 sensor state change	Dashboard/serial monitor evidence captured
PT-05	Python alert communication	Terminal log evidence captured

Table 3.5: Footboard safety system test results

3.4 Subsystem 4 Results: Edge-AI Biometric and Emergency Management

The Edge-AI Facial Recognition and Emergency Management subsystem was evaluated through comprehensive biometric inference testing, edge-computing performance analysis, emergency management workflow validation, and dynamic routing optimization experiments.

The fine-tuned MobileFaceNet convolutional neural network achieved the highest testing accuracy of 99.12% while simultaneously producing the lowest average inference latency of 120 milliseconds on the NVIDIA Jetson Nano edge-computing platform. Comparative evaluation against alternative facial recognition architectures demonstrated that MobileFaceNet significantly outperformed larger models in terms of edge-device efficiency while maintaining excellent biometric identification accuracy.

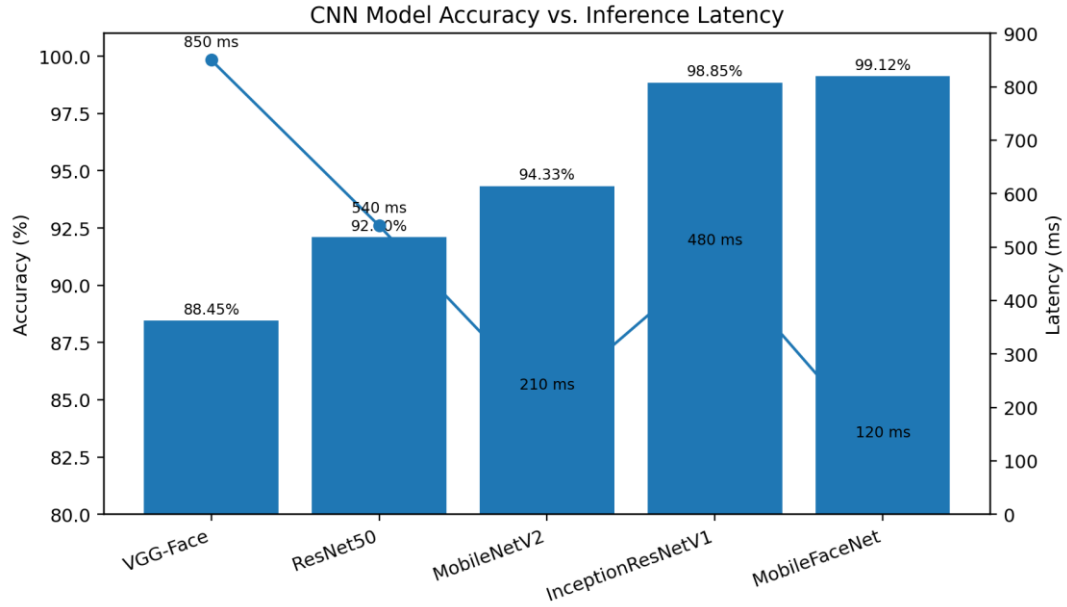


Figure 3.4: CNN model accuracy vs. inference latency comparison

The integrated MTCNN facial detection pipeline successfully localized facial regions under varying passenger positions and moderate illumination variation conditions. Combined preprocessing and inference operations remained sufficiently lightweight for real-time onboard deployment without requiring external cloud-processing infrastructure.

A major focus of subsystem evaluation involved validating decentralized privacy-preserving biometric inference capabilities. Because all recognition operations were performed locally on edge hardware, biometric verification remained operational even during simulated internet outages. This confirmed the effectiveness of the decentralized Edge-AI architecture for maintaining operational continuity while minimizing cloud dependency and bandwidth consumption [38]-[40].

The Context-Aware Automated Emergency Egress System (CA-AEES) was additionally evaluated under simulated emergency conditions involving accident-triggered evacuation scenarios. The subsystem successfully initiated emergency response

workflows and coordinated emergency-control operations following simulated hazard events generated through the integrated accident detection framework.

The Attendance-Driven Dynamic Routing engine was evaluated using multiple transportation route simulations involving varying student attendance patterns. Experimental comparisons between conventional static routing and attendance-aware dynamic routing demonstrated measurable reductions in unnecessary route traversal distances. Through graph-pruning optimization techniques, the routing engine reduced aggregate daily route lengths by up to 11.4%, thereby confirming the subsystem’s ability to reduce ghost-stop inefficiencies and transportation fuel wastage [6], [31], [43], [44].

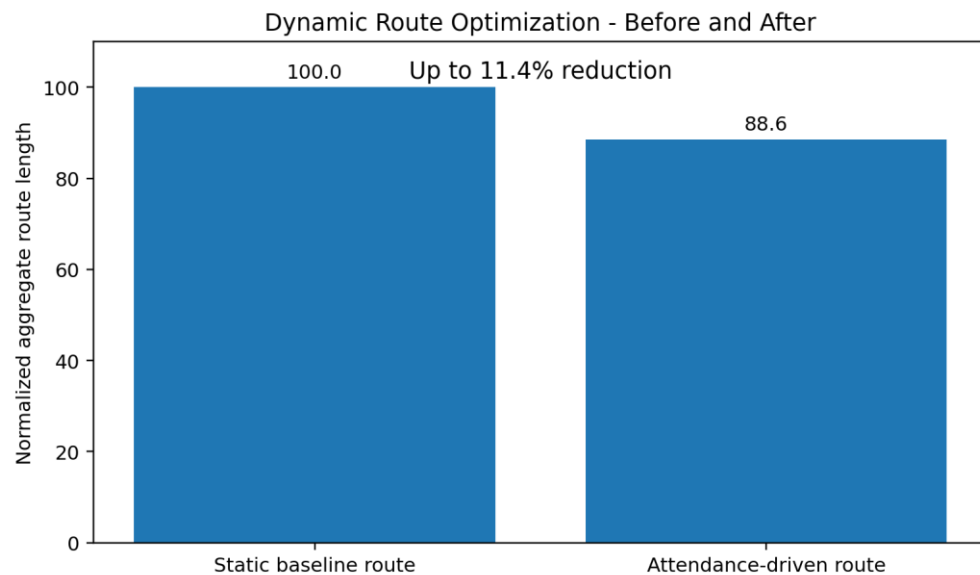


Figure 3.5: Dynamic route optimization

The overall evaluation confirmed that the subsystem successfully achieved high-accuracy biometric verification, low-latency edge inference, privacy-preserving decentralized processing, automated emergency coordination, and intelligent attendance-driven transportation optimization suitable for real-world school transportation deployment.

CNN Architecture	Finalized Accuracy	Average Inference Latency	Selection Outcome
VGG-Face (baseline)	88.45%	850 ms	Rejected due to low accuracy and high latency
ResNet50	92.10%	540 ms	Rejected due to slower inference
MobileNetV2	94.33%	210 ms	Rejected due to lower accuracy
InceptionResNetV1 (standard FaceNet)	98.85%	480 ms	Rejected due to high computational load
MobileFaceNet	99.12%	120 ms	Selected for local SISURAKSHA biometric inference

Table 3.6: CNN architecture accuracy and latency comparison

3.5 Research Findings Summary

The collective experimental results across all four SISURAKSHA subsystems support several significant research findings related to intelligent transportation safety, blockchain verification, IoT reliability, Edge-AI deployment, and operational route optimization.

Finding 1 (Akther): The hybrid blockchain architecture integrating SHA-256 cryptographic hash chaining with Ethereum anchoring provides tamper detection capabilities equivalent to fully on-chain storage architectures while significantly reducing blockchain transaction costs and scalability limitations [27]-[29].

Finding 2 (Akther): Segmented polyline analysis using OSRM-generated road-network geometries substantially improves driver-student assignment realism and route accuracy compared to traditional straight-line Euclidean distance approaches [31]-[34].

Finding 3 (Shameera): Multi-sensor fusion combined with a 60-second driver verification and cancellation mechanism provides a practical and reliable approach for reducing false emergency alerts while maintaining effective accident and hazard detection responsiveness [16], [24].

Finding 4 (Pinto): Hall effect pulse-based movement analysis provides a stable and GPS-independent method for binary vehicle movement classification suitable for safety-critical transportation applications where satellite-based estimation may be unreliable [18], [23].

Finding 5 (Kumara): The fine-tuned MobileFaceNet convolutional neural network achieved 99.12% identification accuracy with an average inference latency of only 120 milliseconds on the NVIDIA Jetson Nano edge device, demonstrating the feasibility of real-time decentralized biometric verification within school transportation environments [40].

Finding 6 (Kumara): The Attendance-Driven Dynamic Routing engine successfully reduced aggregate daily transportation route lengths by up to 11.4% through attendance-aware graph pruning and dynamic route optimization, thereby reducing ghost-stop inefficiencies and operational fuel wastage [6], [31], [43], [44].

3.6 Discussion

The SISURAKSHA platform represents a technically robust, practically relevant, and socially significant contribution to the field of intelligent school transportation systems. By integrating four specialized subsystems addressing complementary dimensions of the school bus safety challenge, the platform achieves a level of operational comprehensiveness that existing isolated transportation solutions generally fail to provide. Rather than treating attendance monitoring, accident detection, passenger safety analysis, biometric verification, and route optimization as independent technological problems, SISURAKSHA demonstrates the feasibility of integrating these capabilities

into a unified cyber-physical transportation ecosystem tailored specifically for the Sri Lankan educational transportation context.

The experimental findings indicate that the platform successfully addresses several critical limitations identified throughout the literature review and research gap analysis phases. Existing school transportation systems within Sri Lanka and many comparable developing regions continue to depend heavily on fragmented manual procedures, reactive communication workflows, and limited GPS-based monitoring infrastructures [1], [3], [4], [6], [31]. Such systems often lack tamper-resistant attendance verification, proactive hazard detection, intelligent emergency management, and transportation efficiency optimization. The SISURAKSHA platform directly responds to these deficiencies through the coordinated integration of blockchain technologies, Edge-AI inference, embedded IoT architectures, computer vision systems, and intelligent routing algorithms.

One of the most significant contributions of the research lies in the hybrid blockchain attendance architecture developed within Subsystem 1. Conventional attendance systems rely predominantly on centralized databases that remain vulnerable to unauthorized modification, synchronization inconsistencies, and insider manipulation [25], [26]. By combining SHA-256 cryptographic hash chaining with lightweight Ethereum anchoring, the proposed subsystem introduces a scalable and economically practical approach for tamper-evident attendance verification. Unlike fully decentralized blockchain architectures that incur high transaction costs and scalability limitations, the hybrid approach demonstrated that immutable integrity verification can be achieved while maintaining operational affordability and synchronization efficiency [27]-[29]. The successful implementation of offline-first MicroSD buffering further strengthens the practical applicability of the architecture within mobile IoT environments characterized by unstable network connectivity.

The segmented polyline route-matching methodology additionally represents an important operational contribution. Traditional transportation assignment systems commonly depend on Euclidean straight-line distance approximations that fail to reflect real-world road network conditions [31]-[34]. The experimental findings demonstrated that OSRM-based segmented polyline analysis significantly improves driver-student assignment realism by evaluating actual navigable transportation geometries. This improvement contributes not only toward operational efficiency but also toward more practical and geographically accurate transportation planning.

Subsystem 2 demonstrated the effectiveness of combining multi-sensor fusion with computer vision-based passenger safety monitoring within transportation environments. Existing accident detection systems often rely on isolated sensor triggers that are highly susceptible to false positives generated by potholes, abrupt braking, or non-critical environmental disturbances [16], [24]. The incorporation of correlated multi-sensor verification combined with a 60-second driver cancellation mechanism significantly improved emergency alert reliability while reducing unnecessary escalation events. This finding is particularly important because excessive false alarms can reduce user trust and operational acceptance of automated safety systems.

The YOLOv8-based passenger window monitoring architecture further illustrates the growing importance of Edge-AI computer vision systems within transportation safety applications. Dangerous passenger behavior such as leaning outside moving buses remains a major safety concern, particularly among school-aged passengers. The ability of the system to perform real-time onboard behavioral analysis demonstrates the feasibility of deploying embedded machine learning models for proactive passenger risk monitoring without requiring continuous cloud-processing infrastructure.

Subsystem 3 contributed important findings regarding movement-aware footboard safety classification. The Hall effect pulse-based movement detection mechanism proved highly reliable for binary movement classification while eliminating the dependence on

unstable GPS-based motion estimation methods [18], [23]. This contribution is operationally significant because GPS signal degradation and positional latency frequently limit the effectiveness of safety-critical transportation systems. The contextual SAFE-WARNING-CRITICAL classification framework additionally demonstrates the importance of integrating occupancy detection with real-time movement analysis rather than relying solely on static occupancy monitoring approaches.

Subsystem 4 produced several notable contributions within the domains of Edge-AI biometric inference, privacy-preserving transportation authentication, and attendance-driven route optimization. The fine-tuned MobileFaceNet architecture achieved exceptionally high biometric identification accuracy while maintaining low inference latency on the NVIDIA Jetson Nano edge platform. This result confirms the suitability of lightweight convolutional neural network architectures for real-time transportation deployment scenarios where computational efficiency and low power consumption are essential operational constraints.

The decentralized biometric inference model additionally represents a meaningful contribution from both technical and regulatory perspectives. Many existing biometric systems rely heavily on centralized cloud infrastructures that introduce bandwidth dependency, increased latency, and privacy concerns associated with sensitive biometric data transmission [38]-[40]. By performing all inference operations locally on edge hardware, SISURAKSHA demonstrates a privacy-preserving architecture more compatible with the requirements of Sri Lanka's Personal Data Protection Act (PDPA) No. 9 of 2022. This design approach reduces external exposure risks while improving operational continuity during internet outages.

Another important contribution of Subsystem 4 involves the Attendance-Driven Dynamic Routing engine. Experimental evaluation demonstrated measurable reductions in route traversal distances through graph-pruning optimization based on verified

attendance data. The observed route-length reduction of up to 11.4% indicates substantial potential for reducing ghost-stop inefficiencies, fuel consumption, transportation costs, and unnecessary environmental emissions [6], [31], [43], [44]. This finding highlights how attendance intelligence can extend beyond administrative recordkeeping and directly contribute toward operational transportation optimization.

Collectively, the four integrated subsystems demonstrate the practical feasibility of constructing an intelligent transportation ecosystem capable of simultaneously addressing security, safety, accountability, operational efficiency, and emergency responsiveness within school transportation environments. The modular system architecture additionally improves scalability and future extensibility because individual subsystems may be independently upgraded, deployed, or expanded without disrupting the entire ecosystem.

Despite these strengths, several limitations remain within the current implementation. Certain experimental evaluations were conducted within controlled or semi-controlled environments rather than full-scale long-duration real-world deployments. Consequently, additional large-scale field testing involving operational school bus fleets would further strengthen empirical validation under diverse environmental and operational conditions. Similarly, while the blockchain anchoring architecture significantly reduces operational costs compared to fully on-chain systems, long-term blockchain scalability and transaction fee variability remain practical considerations for nationwide deployment.

The computer vision subsystem may also experience performance variability under extreme lighting conditions, severe occlusion scenarios, or highly crowded passenger environments. Additionally, while MobileFaceNet demonstrated excellent inference efficiency, future research may explore federated learning approaches, multimodal biometric verification, or transformer-based lightweight architectures to further improve recognition robustness and adaptability.

Future extensions of the SISURAKSHA ecosystem may include predictive maintenance analytics, AI-driven driver fatigue monitoring, advanced evacuation simulation systems, autonomous emergency communication with public authorities, vehicle-to-infrastructure (V2I) communication, and integration with broader smart-city transportation infrastructures. Additional work may also investigate reinforcement learning-based route optimization models capable of adapting dynamically to traffic congestion, weather conditions, and evolving attendance patterns.

Overall, the findings of this research confirm that intelligent school transportation systems integrating blockchain technologies, IoT architectures, Edge-AI inference, computer vision monitoring, and dynamic route optimization can substantially improve child transportation safety, operational accountability, and transportation efficiency. The SISURAKSHA platform therefore represents both a meaningful academic contribution and a practically deployable foundation for future smart educational transportation ecosystems within Sri Lanka and similar developing transportation environments [1], [4], [6], [25], [31], [38].

CHAPTER 4: SUMMARY OF EACH STUDENT'S CONTRIBUTION

The SISURAKSHA project was developed collaboratively by four undergraduate students, each responsible for a distinct but interoperable subsystem.

Team Member	Subsystem / Component	Key Contributions
M. H. S. Akther (IT22590312)	Blockchain-Based RFID Attendance and Segmented Polyline Analysis	IoT Smart Attendance Device, hybrid blockchain architecture, SHA-256 hash chaining, segmented polyline algorithm, three-layer blockchain validation framework
T. A. P. K. Shameera (IT22606624)	Smart Bus Crash Detection and Safety Window Monitoring	Multi-sensor accident and hazard detection, 60-second driver verification mechanism, YOLOv8-based window safety monitoring, GSM emergency alert dispatch
P. R. I. S. R. Pinto (IT22610102)	ML and IoT Footboard Safety and Driver Monitoring	IR sensor footboard occupancy detection, Hall effect pulse-based bus movement detection, three-state risk classification, LM2596 power supply design
A. M. D. C. Kumara (IT22600134)	Smart Access and Emergency Management with Edge-AI Facial Recognition	Fine-tuned MobileFaceNet, MTCNN facial detection pipeline, AES-256 encrypted biometric template storage, CA-AEES dual-verification interlock, Attendance-Driven Dynamic Routing engine

Table 3.7: Summary of each student's contribution

4.1 M. H. S. Akther (IT22590312)

Akther's contribution constitutes the foundational trust and intelligence layer of the SISURAKSHA ecosystem. The design and implementation of the custom IoT Smart

Attendance Device from hardware selection through firmware development to testing demonstrated a comprehensive command of embedded systems engineering.

4.2 T. A. P. K. Shameera (IT22606624)

Shameera's contribution addresses the physical safety and environmental hazard monitoring dimensions of the SISURAKSHA platform. The design and integration of a comprehensive multi-sensor array covering accelerometric, vibration, pressure, fire, gas, and water hazard detection into a single ESP32-based system demonstrates strong hardware integration competence.

4.3 P. R. I. S. R. Pinto (IT22610102)

Pinto's contribution focuses on the footboard safety monitoring problem, which represents one of the highest-risk zones in school bus operations and has received limited attention in existing research. The development and validation of Hall effect pulse-based movement detection as an alternative to GPS-based speed measurement represents a key methodological finding of this subsystem.

4.4 A. M. D. C. Kumara (IT22600134)

Kumara's contribution addresses the biometric accountability, data privacy, and route optimization dimensions of the SISURAKSHA platform. The systematic comparative evaluation of five CNN architectures for edge-device biometric classification, resulting in the selection and fine-tuning of MobileFaceNet to achieve 99.12% accuracy at 120 ms latency, constitutes a rigorous machine learning engineering contribution.

CHAPTER 5: CONCLUSIONS

5.1 Summary of Research

This dissertation presented SISURAKSHA, an Intelligent Child Safety Platform for School Buses, developed as a collaborative undergraduate research project aimed at addressing the critical and increasingly complex safety, accountability, and operational efficiency challenges associated with school transportation systems in Sri Lanka. The research was motivated by the growing number of transportation-related risks affecting school children, the operational inefficiencies of conventional transportation management systems, and the absence of integrated technological ecosystems capable of providing proactive transportation safety management [1], [3], [4].

The proposed SISURAKSHA platform was conceptualized and implemented as a modular cyber-physical ecosystem integrating four specialized but interoperable subsystems. These include a blockchain-based RFID attendance management and segmented polyline route-matching subsystem; a multi-sensor crash detection and YOLOv8-based passenger window safety monitoring subsystem; a Hall effect movement-aware footboard safety monitoring subsystem; and an Edge-AI biometric authentication, attendance-driven dynamic routing, and automated emergency management subsystem. Collectively, these subsystems address nine major research gaps identified during the literature review and problem analysis phases.

The first subsystem introduced a hybrid blockchain attendance architecture integrating RFID-based student verification, offline-first IoT synchronization, SHA-256 cryptographic hash chaining, and Ethereum blockchain anchoring. The subsystem additionally incorporated segmented polyline analysis using OSRM-generated road-network geometries to improve driver-student route assignment realism and transportation efficiency [25]-[29], [31]-[34].

The second subsystem focused on transportation accident detection and passenger safety monitoring through the integration of multi-sensor IoT architectures and computer vision technologies. An ESP32-based sensor-fusion framework was developed to detect collisions, rollovers, smoke, gas leakage, fire hazards, water intrusion, and abnormal vibration conditions while reducing false alerts through a driver verification mechanism. In parallel, a YOLOv8-based machine learning model was implemented to identify unsafe passenger window behavior in real time [16], [19], [20], [24].

The third subsystem addressed the high-risk operational challenges associated with school bus footboard and entrance areas. By integrating Hall effect pulse-based movement detection with infrared occupancy sensing, the subsystem implemented a movement-aware contextual risk classification model capable of distinguishing between safe and hazardous passenger occupancy conditions during vehicle operation [18], [22], [23].

The fourth subsystem introduced a decentralized Edge-AI biometric authentication architecture using MobileFaceNet and MTCNN deployed on NVIDIA Jetson Nano hardware. The subsystem additionally incorporated an Attendance-Driven Dynamic Routing engine and a Context-Aware Automated Emergency Egress System designed to improve transportation efficiency and emergency responsiveness while maintaining compliance with Sri Lanka's Personal Data Protection Act (PDPA) No. 9 of 2022 [38]-[40], [43], [44].

Experimental evaluation across all four subsystems demonstrated strong operational performance, high reliability, low inference latency, robust offline resilience, and practical feasibility for deployment within real-world transportation environments. The collective findings confirm that intelligent school transportation ecosystems integrating blockchain architectures, Edge-AI computing, embedded IoT systems, computer vision technologies, and dynamic routing algorithms can substantially improve child safety,

operational transparency, emergency responsiveness, and transportation efficiency within educational transportation systems.

5.2 Achievement of Research Objectives

The research successfully achieved all group-level and individual subsystem objectives established during the problem formulation and system design phases. Each subsystem fulfilled its intended operational goals while collectively contributing toward the realization of a unified intelligent school transportation ecosystem.

The objective of developing a reliable IoT-based RFID attendance management system was successfully achieved through the implementation of the ESP32-based Smart Attendance Device integrated with the PN532 RFID reader module and offline-first MicroSD persistence architecture. Experimental evaluation confirmed a 100% RFID read success rate across 200 attendance scan simulations while the deferred synchronization mechanism achieved a 99.5% synchronization success rate under simulated network interruption conditions [8], [9], [15], [30].

The objective of establishing tamper-resistant attendance verification was achieved through the successful implementation of a hybrid blockchain architecture combining MongoDB attendance blocks, PostgreSQL operational data management, SHA-256 cryptographic hash chaining, and Ethereum blockchain anchoring. Experimental tampering simulations confirmed that unauthorized attendance modification attempts immediately invalidated blockchain continuity, thereby demonstrating effective tamper detection capabilities equivalent to full on-chain verification approaches while significantly reducing transaction costs [27]-[29].

The route intelligence objective was achieved through the implementation of segmented polyline analysis using OSRM-generated road-network geometries. Experimental evaluations demonstrated that the proposed route-matching methodology produced more

realistic and geographically accurate driver-student assignments than traditional straight-line Euclidean distance approaches [31]-[34].

The accident and hazard detection objective was successfully fulfilled through the development of the ESP32-based multi-sensor crash detection architecture integrating accelerometric, vibration, smoke, gas, water, and pressure-monitoring sensors. Experimental evaluation confirmed reliable hazard recognition capabilities while the 60-second driver verification and cancellation mechanism effectively reduced false emergency escalations [16], [17], [24].

The objective of implementing real-time passenger safety monitoring was achieved through the deployment of the YOLOv8-based computer vision framework capable of identifying unsafe passenger window behavior under real-time operational conditions. The subsystem successfully demonstrated embedded behavioral analysis suitable for proactive transportation safety monitoring.

The footboard safety objective was achieved through the implementation of the Hall effect pulse-based movement detection architecture integrated with infrared occupancy sensing. Experimental testing confirmed reliable movement-aware contextual classification across SAFE, WARNING, and CRITICAL operational states, thereby validating the feasibility of GPS-independent transportation movement analysis [18], [22], [23].

The Edge-AI biometric authentication objective was successfully accomplished through the implementation of the MobileFaceNet and MTCNN-based decentralized facial recognition architecture deployed on the NVIDIA Jetson Nano platform. The fine-tuned MobileFaceNet model achieved 99.12% identification accuracy with an average inference latency of 120 milliseconds, thereby satisfying the operational requirements for real-time onboard biometric verification [40].

The dynamic routing optimization objective was achieved through the implementation of the Attendance-Driven Dynamic Routing engine capable of pruning unnecessary transportation nodes associated with absent students. Experimental results demonstrated route-length reductions of up to 11.4%, confirming the subsystem's ability to reduce ghost-stop inefficiencies and transportation fuel wastage [6], [31], [43], [44].

Finally, the overall group objective of constructing an interoperable intelligent transportation ecosystem was achieved through the successful integration of all four subsystems into a cohesive modular platform capable of simultaneously addressing attendance verification, transportation safety monitoring, emergency management, passenger behavior analysis, biometric authentication, and transportation optimization.

5.3 Conclusions

The SISURAKSHA platform demonstrates that a unified, modular, and interoperable cyber-physical ecosystem can effectively address the multifaceted challenges associated with school transportation safety, accountability, emergency responsiveness, and operational efficiency within the Sri Lankan context. By integrating IoT edge computing, cryptographic blockchain verification, machine learning-based safety monitoring, biometric identification, embedded computer vision, and intelligent transportation optimization into a single cohesive architecture, the platform transforms the traditional school transportation environment from a reactive and compliance-oriented operational model into a proactive, autonomous, and data-driven safety ecosystem.

The research confirms that blockchain technologies can be practically integrated within transportation attendance systems through hybrid architectures that balance integrity verification, scalability, and operational affordability. The successful implementation of SHA-256 cryptographic hash chaining combined with lightweight Ethereum anchoring demonstrates that tamper-resistant attendance verification can be achieved without

incurring the prohibitive operational costs associated with fully decentralized blockchain infrastructures [27]-[29].

The findings additionally demonstrate the practical feasibility of deploying multi-sensor IoT architectures and Edge-AI computer vision systems within transportation environments for proactive hazard detection and passenger safety monitoring. The integration of sensor fusion techniques with machine learning inference significantly improves transportation situational awareness and emergency responsiveness while reducing dependence on manual supervision and delayed communication workflows [16], [19], [20], [24].

The Hall effect movement-aware footboard safety subsystem further illustrates the importance of contextual safety analysis within transportation systems. By integrating occupancy detection with real-time motion analysis, the research demonstrates a practical alternative to unreliable GPS-dependent movement estimation approaches commonly utilized within conventional transportation monitoring systems [18], [22], [23].

The Edge-AI biometric subsystem confirms that decentralized facial recognition systems can achieve high identification accuracy and low inference latency while preserving privacy and minimizing cloud dependency. This finding is particularly important within regulatory-sensitive educational environments involving minors and sensitive biometric information governed by Sri Lanka's Personal Data Protection Act (PDPA) No. 9 of 2022 [38]-[40].

The Attendance-Driven Dynamic Routing engine additionally demonstrates how transportation intelligence can extend beyond administrative attendance management toward measurable operational optimization. By dynamically pruning unnecessary route nodes associated with absent students, the platform reduces ghost-stop inefficiencies, transportation fuel consumption, and operational overhead while improving route efficiency [6], [31], [43], [44].

Overall, the SISURAKSHA research contributes significantly to the fields of intelligent transportation systems, IoT-enabled safety architectures, Edge-AI computing, blockchain-based verification systems, and smart educational infrastructure development. The platform not only provides a technically viable solution for improving school transportation safety within Sri Lanka, but also establishes a scalable conceptual foundation for future intelligent transportation ecosystems applicable to broader developing-world educational transportation contexts.

The research therefore concludes that integrated intelligent transportation ecosystems such as SISURAKSHA possess substantial potential to improve child safety, operational accountability, transportation transparency, and emergency responsiveness while supporting the long-term modernization of educational transportation infrastructure through emerging digital technologies [1], [4], [25], [31], [38].

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